MATH READING NOTES: *Harmonic Analysis: Oscillatory Integrals of the First Kind*

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Based on the book *Harmonic Analysis: Real-Variable Methods, Orthogonality, and Oscillatory Integrals* by Elias M. Stein

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Disclaimer: *These notes are informal, and have not been peer-reviewed, which may contain errors. Feedback is welcome (contact the author via email).* ♟🐰

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1 Overview

1.1 Oscillatory Integrals: Key Concepts and Asymptotics

In this chapter, we discuss oscillatory integrals of the first kind:

$$I(\lambda) = \int_a^b e^{i\lambda\phi(x)}\psi(x)\,dx. \tag{1.1}$$

where $\lambda > 0$ (*frequency*), ϕ is a real-valued smooth function (*phase function*), and ψ is a complex-valued smooth function (*amplitude/envelope function*) with compact support in (a, b).

Physically, these terms can be understood as:

i. λ (*frequency*): it's a large parameter controlling oscillation frequency.

ii. $\phi(x)$ (*phase function*): it governs the frequency and location of oscillations (via its derivatives).

iii. $\psi(x)$ (*amplitude function*): it determines the "weight" of the oscillations at each point x.

iv. $e^{i\lambda\phi(x)} = \cos(\lambda\phi(x)) + i\sin(\lambda\phi(x))$ is an *oscillatory kernel*, where $\lambda\phi(x)$ is its *total phase*.

Note: $\phi(x)$ (*phase function*) determines where oscillations slow down at critical points, whereas $\lambda\phi(x)$ (*total phase*) controls *frequency* of oscillations. The imaginary i ensures the integrand is oscillatory, enabling destructive/constructive interference.

To understand the behavior of oscillatory integrals, we use asymptotic analysis. Because exact solutions are often impossible or impractical, especially when $\lambda \to \infty$.

Asymptotic principles of $I(\lambda)$ as $\lambda \to \infty$:

i. **Contributions from critical points**: The dominant contributions arise from the critical points of ϕ (where $\phi'(x) = 0$).

ii. **Quantitative dependence on degeneracy**: The asymptotic behavior is determined by the order of vanishing of ϕ' at the critical points.

iii. **Universal decay estimates**: Lower bounds on certain derivatives of ϕ yield general decay estimates.

Methodology for asymptotics:

i. **Localization**: Restrict the integral to neighborhoods of critical points.

ii. **Scaling**: Rescale the integral to magnify the behavior near critical points.

iii. **Asymptotics**: Apply stationary phase or other methods to extract the leading-order behavior.

2 OSCILLATORY INTEGRALS OF THE FIRST KIND, ONE VARIABLE

2.1 Localization

Intuition: When $\lambda \gg 1$, the integrand $e^{i\lambda\phi(x)}\psi(x)$ oscillates rapidly, causing destructive interference everywhere except near:

i. Critical points for ϕ (where $\phi'(x) = 0$), where oscillations "slow down" and constructive interference may occur.

ii. Boundary points that survive (if ψ does not vanish there), though compact support often eliminates these.

Thus, we can:

1. Localize the integral to small neighborhoods of critical points (where $\phi'(x) = 0$).
2. Discard regions where $\phi'(x) \neq 0$, since repeated integration by parts shows that they contribute $\mathcal{O}(\lambda^{-N})$ for any $N > 0$.

[Proposition 1: Non-Stationary Phase Decay] Let:

i. $\phi \in C^\infty([a,b])$ with $\phi'(x) \neq 0$ for all $x \in [a,b]$;

ii. $\psi \in C_c^\infty((a,b))$ (smooth and compactly supported).

Then, the oscillatory integral satisfies:

$$\forall N \geq 0, \quad I(\lambda) := \int_a^b e^{i\lambda\phi(x)}\psi(x)\,dx = \mathcal{O}(\lambda^{-N}) \quad \text{as } \lambda \to \infty. \tag{2.1}$$

Proof.

1. *Step 1: Rewrite the oscillatory factor:* We want to isolate the oscillation's derivative, by expressing $e^{i\lambda\phi(x)}$ in terms of its own derivative. This links the oscillatory part to the phase function's derivative $\phi'(x)$. It helps to control the local frequency of oscillations.

   Since $\phi'(x) \neq 0$, we have:

$$\frac{d}{dx}\left(e^{i\lambda\phi(x)}\right) = i\lambda\phi'(x) \implies e^{i\lambda\phi(x)} = (i\lambda\phi'(x))^{-1}\frac{d}{dx}\left(e^{i\lambda\phi(x)}\right). \tag{2.2}$$

2. *Step 2: Integration by parts once:* IBP moves the derivative from the oscillatory factor $e^{i\lambda\phi(x)}$ to the amplitude $\psi(x)/\phi'(x)$. So, each IBP "trades" oscillation strength λ for smoothness (derivatives of ψ/ϕ').

$$I(\lambda) = \int_a^b e^{i\lambda\phi(x)}\psi(x)dx = \int_a^b (i\lambda\phi'(x))^{-1}\frac{d}{dx}\left(e^{i\lambda\phi(x)}\right)\psi(x)dx. \tag{2.3}$$

let $u(x) = \psi(x)(i\lambda\phi'(x))^{-1}$, $dv(x) = d/dx\left(e^{i\lambda\phi(x)}\right)dx$

$du(x) = d/dx\left(\psi(x)(i\lambda\phi'(x))^{-1}\right)dx$, $v(x) = e^{i\lambda\phi(x)}$:

$$I(\lambda) = \Big[uv\Big] - \int v\,du = \underbrace{\left[\frac{\psi(x)e^{i\lambda\phi(x)}}{i\lambda\phi'(x)}\right]_a^b}_{\text{since } \psi(a)=\psi(b)=0} - \int_a^b e^{i\lambda\phi(x)}\frac{d}{dx}\left(\psi(x)(i\lambda\phi'(x))^{-1}\right)dx.$$

$$= (i\lambda)^{-1}\int_a^b e^{i\lambda\phi(x)}\underbrace{\left(-\frac{d}{dx}\left(\psi(x)(\phi'(x))^{-1}\right)\right)}_{:=\Psi_1(x)\in C_c^\infty}dx = (i\lambda)^{-1}\int_a^b e^{i\lambda\phi(x)}\Psi_1(x)\,dx. \quad (2.4)$$

3. *Step 3: Integration by parts N times:*

Repeated IBP shows $I(\lambda)$ dcays faster than any polynomial in λ (non-stationary phase). It also shows that how the amplitude's smoothness ($\Psi_N \in C_c^\infty$) "tames" the oscillations.

[**Note:** the negative sign can be absorbed into the definition of $\Psi(x)$ purely for notational convenience. Because the decay rate $\mathcal{O}(\lambda^{-N})$ depends only on the magnitude of $(i\lambda)^{-N}$, not the sign.]

After N repetitions of IBP, we have:

$$I(\lambda) = (i\lambda)^{-1}\int_a^b e^{i\lambda\phi(x)}\underbrace{\Psi_1(x)}_{\Psi_1\in C_c^\infty}dx \implies I(\lambda) = (i\lambda)^{-N}\int_a^b e^{i\lambda\phi(x)}\underbrace{\Psi_N(x)}_{\Psi_N\in C_c^\infty}dx. \quad (2.5)$$

where Ψ_N is defined recursively with:

$$\Psi_k(x) = -\frac{d}{dx}\left(\Psi_{k-1}(x)(\phi'(x))^{-1}\right), \quad \Psi_0(x) = \psi(x). \quad (2.6)$$

Since $\phi'(x) \neq 0$ and $\psi \in C_c^\infty$, each Ψ_N remains smooth and compactly supported.

To bound $I(\lambda)$, through the triangle inequality for integrals, we observe that:

$$\left|\int f(x)dx\right| \leq \int |f(x)|dx \implies \left|\int_a^b e^{i\lambda\phi(x)}\Psi_N(x)\,dx\right| \leq \int_a^b |\Psi_N(x)|dx. \quad (2.7)$$

where $|e^{i\lambda\phi(x)}| = 1$ for all x based on Euler's formula. Thus we have:

$$|I(\lambda)| \leq \lambda^{-N}\int_a^b \underbrace{|\Psi_N(x)|}_{C_N}dx \leq C_N\lambda^{-N} = \mathcal{O}(\lambda^{-N}) \quad \text{as } \lambda\to\infty. \quad (2.8)$$

where C_N is a constant independent of λ. Moreover, the above equation holds for any $N \geq 0$, proving the rapid decay claim.

□

Note: In Stein's notation, the proof remains the same but with a definition difference, where he uses the formal adjoint ${}^tD$ via L^2 pairing. We can check this as below.

Proof.

1. *Step 1: Rewrite the oscillatory factor:* We do the following definitions:

$$D(f(x)) = (i\lambda\phi'(x))^{-1}df/dx \implies D\left(e^{i\lambda\phi(x)}\right) = (i\lambda\phi'(x))^{-1}\frac{d}{dx}\left(e^{i\lambda\phi(x)}\right). \tag{2.9}$$

$${}^tD(g(x)) = -d/dx\left(g/i\lambda\phi'(x)\right) \implies {}^tD(g(x)) = -\frac{d}{dx}\left(\psi(x)(i\lambda\phi(x))^{-1}\right). \tag{2.10}$$

where $f(x) = e^{i\lambda\phi(x)}$ and $g(x) = \psi(x)$.

These satisfy the L^2 adjoint identity, with vanishing boundary terms for $f, g \in C_c^\infty$:

$$\int_a^b (Df(x))g(x)dx = \int_a^b f(x)({}^tDg(x))dx. \tag{2.11}$$

where ${}^tD$ is the formal adjoint (transpose) of D.

2. *Step 2: Integration by parts:* Here, we do IBP by applying D to the oscillatory factor:

$$e^{i\lambda\phi(x)} = D\left(e^{i\lambda\phi(x)}\right). \tag{2.12}$$

Substitute this into $I(\lambda)$, which gives:

$$\begin{aligned} I(\lambda) &= \int_a^b e^{i\lambda\phi(x)}\psi(x)dx = \int_a^b D(e^{i\lambda\phi(x)})\psi(x)dx. \\ &= \int_a^b \underbrace{(i\lambda\phi'(x))^{-1}\frac{d}{dx}\left(e^{i\lambda\phi(x)}\right)}_{D(f(x)) = D(e^{i\lambda\phi(x)})}\psi(x)dx = \int_a^b e^{i\lambda\phi(x)}\underbrace{\left[-\frac{d}{dx}\left(\psi(x)(i\lambda\phi(x))^{-1}\right)\right]}_{{}^tD(g(x))={}^tD(\psi(x))}dx. \end{aligned} \tag{2.13}$$

Everytime, we do integration by parts, we replace:

$$\int D(e^{i\lambda\phi(x)})\psi(x)dx \quad \text{with} \quad \int e^{i\lambda\phi(x)}({}^tD\psi(x))dx. \tag{2.14}$$

We do it for N times, we then have:

$$I(\lambda) = \int_a^b e^{i\lambda\phi(x)}\psi(x)dx \implies I(\lambda) = \int_a^b D^N(e^{i\lambda\phi(x)})\psi(x)dx = \int_a^b e^{i\lambda\phi(x)}\cdot\left({}^tD\right)^N(\psi(x))dx. \tag{2.15}$$

where $({}^tD)^N\psi$ is the N-th iterate of ${}^tD$ on ψ. Since $\psi \in C_c^\infty$ and $\phi' \neq 0$, each $({}^tD)^N\psi$ remains smooth and compactly supported. We thus have the same result:

$$|I(\lambda)| \leq \lambda^{-N}\int_a^b |({}^tD)^N\psi(x)|\,dx = \mathcal{O}(\lambda^{-N}). \tag{2.16}$$

□

2.1.1 Edge Case i: Non-Vanishing Boundary Terms

If $\psi(x)$ does not vanish at $x = a$ or $x = b$, we then have the boundary terms. Let:

$$I(\lambda) = \int_a^b e^{i\lambda\phi(x)}\psi(x)dx \text{ with } e^{i\lambda\phi(x)} = (i\lambda\phi'(x))^{-1}d/dx\left(e^{i\lambda\phi(x)}\right). \tag{2.17}$$

After one integration by parts:

$$\begin{aligned} I(\lambda) = \int_a^b e^{i\lambda\phi(x)}\psi(x)dx &= \int_a^b (i\lambda\phi(x))^{-1}d/dx\left(e^{i\lambda\phi(x)}\right)\psi(x)dx. \\ &= \underbrace{\left[\frac{\psi(x)\cdot e^{i\lambda\phi(x)}}{i\lambda\phi'(x)}\right]_a^b}_{\text{Boundary terms: } B_1(\lambda)} - \underbrace{\int_a^b e^{i\lambda\phi(x)}\frac{d}{dx}\left(\psi(x)(i\lambda\phi'(x)^{-1})\right)dx}_{\text{Remainder integral: } R_1(\lambda)}. \\ &= B_1(\lambda) + (i\lambda)^{-1}\int_a^b e^{i\lambda\phi(x)}\Psi_1(x)\,dx. \end{aligned} \tag{2.18}$$

We then investigate on the boundary terms $B_1(\lambda)$:

$$\left[\frac{\psi(x)\cdot e^{i\lambda\phi(x)}}{i\lambda\phi'(x)}\right]_a^b = \frac{\psi(b)\cdot e^{i\lambda\phi(b)}}{i\lambda\phi'(b)} - \frac{\psi(a)\cdot e^{i\lambda\phi(a)}}{i\lambda\phi'(a)} \implies \underbrace{(i\lambda)^{-1}\left[\frac{{}^tD^0\psi(b)\cdot e^{i\lambda\phi(b)}}{\phi'(b)} - \frac{{}^tD^0\psi(a)\cdot e^{i\lambda\phi(a)}}{\phi'(a)}\right]}_{\text{Endpoint terms} = \mathcal{O}(\lambda^{-1})}. \tag{2.19}$$

where ${}^tD^0\psi(x) = \psi(x)$ for $k = 1$. So, $B_1(\lambda) = \mathcal{O}(\lambda^{-1})$. Now, repeat this for N times:

$$I(\lambda) = \sum_{k=1}^{N} B_k(\lambda) + R_N(\lambda). \tag{2.20}$$

with $B_k(\lambda) = \mathcal{O}(\lambda^{-k})$, where $({}^tD)^{k-1}$ acts on $\psi(x)$ before evaluation at $x = a, b$:

$$B_k(\lambda) = \left[\frac{({}^tD)^{k-1}\psi(x)\cdot e^{i\lambda\phi(x)}}{(i\lambda)^k(\phi'(x))^k}\right]_a^b \text{ with } {}^tD\psi(x) = -\frac{d}{dx}\left(\frac{\psi(x)}{\phi'(x)}\right), {}^tD^k = {}^tD\circ{}^tD^{k-1}. \tag{2.21}$$

and the remainder, including the functions $\Psi_k(x)$ that's defined recursively:

$$R_N(\lambda) = (i\lambda)^{-N}\int_a^b e^{i\lambda\phi(x)}\Psi_N(x)\,dx = \mathcal{O}(\lambda^{-N}) \text{ with } \Psi_k(x) = -\frac{d}{dx}\left(\frac{\Psi_{k-1}(x)}{\phi'(x)}\right), \Psi_0(x) = \psi(x). \tag{2.22}$$

Thus, the asymptotic expansion is:

$$I(\lambda) = \sum_{k=1}^{N}\frac{c_k}{\lambda^k} + \mathcal{O}(\lambda^{-N}) \quad c_k = \frac{(-1)^k}{i^k}\left[\frac{({}^tD)^{k-1}\psi(x)}{\phi'(x)^k}\right]_a^b. \tag{2.23}$$

where c_k depend on ψ, ϕ at $x = a, b$.

Remark: Vanishing vs. Non-Vanishing Amplitude at Endpoints

The asymptotic behavior of $I(\lambda)$ depends critically on whether $\psi(x)$ vanishes at $x = a, b$:

i. **Case A (Proposition 1, ref: 2.1)** $\psi(a) = \psi(b) = 0$ (Vanishing endpoints). All boundary terms vanish, leaving only the remainder integral:

$$I(\lambda) = (i\lambda)^{-N} \int_a^b e^{i\lambda\phi(x)} \Psi_N(x)\, dx = \mathcal{O}(\lambda^{-N}), \quad \forall N \geq 0. \tag{2.24}$$

Each IBP transfers derivatives to $\Psi_N(x)$ while extracting λ^{-1}.

ii. **Case B (Edge Case I):** $\psi(a) \neq 0$ or $\psi(b) \neq 0$ (Non-vanishing endpoints). Boundary terms $B_k(\lambda)$ appear, and the expansion becomes:

$$I(\lambda) = \sum_{k=1}^{N} B_k(\lambda) + R_N(\lambda), \quad B_k(\lambda) = \mathcal{O}(\lambda^{-k}), \quad R_N(\lambda) = \mathcal{O}(\lambda^{-N}). \tag{2.25}$$

Each IBP step:

1. Generates a boundary term $B_k(\lambda) = \mathcal{O}(\lambda^{-k})$.
2. Leaves a remainder integral $R_k(\lambda)$ with λ^{-k} decay.

Mechanism:

$$I(\lambda) = \underbrace{\left[\frac{\psi(x)e^{i\lambda\phi(x)}}{i\lambda\phi(x)}\right]_a^b}_{B_1(\lambda)=\mathcal{O}(\lambda^{-1})} - \underbrace{(i\lambda)^{-1} \int_a^b e^{i\lambda\phi(x)} \Psi_1(x)\, dx}_{R_1(\lambda)}. \tag{2.26}$$

Take the remainder and do the second IBP:

$$\begin{aligned} R_1(\lambda) &= (i\lambda)^{-1} \int_a^b e^{i\lambda\phi(x)} \Psi_1(x)\, \mathrm{d}x. \\ &= \underbrace{(i\lambda)^{-1} \left[\frac{\Psi_1(x)e^{i\lambda\phi(x)}}{i\lambda\,\phi'(x)}\right]_a^b}_{B_2(\lambda)=\mathcal{O}(\lambda^{-1})\cdot\mathcal{O}(\lambda^{-1})=\mathcal{O}(\lambda^{-2})} - \underbrace{(i\lambda)^{-2} \int_a^b e^{i\lambda\phi(x)} \Psi_2(x)\, \mathrm{d}x}_{R_2(\lambda)}. \end{aligned} \tag{2.27}$$

Keep doing IBP for k times, we then have the boundary terms as:

$$B_1(\lambda) = \mathcal{O}(\lambda^{-1}), B_2(\lambda) = \mathcal{O}(\lambda^{-2}), \cdots, B_k(\lambda) = \mathcal{O}(\lambda^{-k}). \tag{2.28}$$

and a remainder integral:

$$R_k(\lambda) = (i\lambda)^{-k} \int e^{i\lambda\phi(x)} \Psi_k(x) dx = \mathcal{O}(\lambda^{-k}). \tag{2.29}$$

Key Insight: Whether ψ vanishes at the endpoints only affects the *explicit boundary contributions* $B_k(\lambda)$. The *remainder integral* $R_N(\lambda)$ always decays as $\mathcal{O}(\lambda^{-N})$ due to the λ^{-1} factor from each IBP.

2.1.2 Edge Case ii: Periodic Boundary Conditions

If $\phi(x)$ and $\psi(x)$ are smooth and periodic on $[a, b]$ with matching derivatives at the endpoints:

$$\phi^{(k)}(a) = \phi^{(k)}(b) \quad \text{and} \quad \psi^{(k)}(a) = \psi^{(k)}(b) \quad \text{for all } k \geq 0. \tag{2.30}$$

Then, integration by parts yields vanishing boundary terms, and the integral $I(\lambda)$ decays rapidly.

Mechanism of Cancellation:

1. First Integration by Parts:

$$I(\lambda) = \underbrace{\left[\frac{\psi(x)e^{i\lambda\phi(x)}}{i\lambda\phi'(x)}\right]_a^b}_{B_1(\lambda)} - (i\lambda)^{-1}\int_a^b e^{i\lambda\phi(x)}\Psi_1(x)\,dx. \tag{2.31}$$

   The boundary term $B_1(\lambda)$ vanishes because:

$$\psi(a)e^{i\lambda\phi(a)} = \psi(b)e^{i\lambda\phi(b)} \quad \text{and} \quad \phi'(a) = \phi'(b). \tag{2.32}$$

2. Higher-Order Terms: For the k-th iteration, the boundary term involves:

$$B_k(\lambda) = \left[\frac{({}^tD)^{k-1}\psi(x)\cdot e^{i\lambda\phi(x)}}{(i\lambda)^k(\phi'(x))^k}\right]_a^b. \tag{2.33}$$

   Periodicity ensures $({}^tD)^{k-1}\psi(a) = ({}^tD)^{k-1}\psi(b)$ and $\phi^{(m)}(a) = \phi^{(m)}(b)$, so all $B_k(\lambda) = 0$.

3. Result:

$$I(\lambda) = (i\lambda)^{-N}\int_a^b e^{i\lambda\phi(x)}\Psi_N(x)\,dx = \mathcal{O}(\lambda^{-N}) \quad \forall N \geq 0. \tag{2.34}$$

All boundary terms vanish due to periodicity. Physically, this means that periodicity causes destructive interference at endpoints, mimicking Case A ($\psi(a) = \psi(b) = 0$).

2.2 Scaling

Intuition: In $\mathbb{R}$, how many times can we effectively differentiate away the oscillations? We consider two scenarios for the oscillatory part ($I(\lambda) = \int_a^b e^{i\lambda\phi(x)}$) in the oscillatory integral:

i. **Non-vanishing first derivative:** If $\phi'(x) \neq 0$, we can differentiate the oscillations infinitely often. The phase $\phi(x)$ is strictly increasing or decreasing — no "flat" regions. Physically, the first derivative $\phi'(x)$ measures the instantaneous rate of change of $\phi(x)$, like velocity.

ii. **Non-vanishing k-th derivative:** If $\phi^{(k)}(x) \neq 0$ (with lower derivatives possibly vanishing), we can gain a decay of order $\lambda^{-1/k}$, but no further improvement is possible. In this case, we measure higher-order curvature (how the $(k-1)$-th derivative changes).

① : For scenario i, it leads to arbitrarily fast decay (e.g., $I(\lambda) \sim \lambda^{-N}$ for any N. Prop 1: (2.1)).
② : For scenario ii, we can do the following:

1. **Assumption:** Suppose the k-th derivative of the phase is bounded below by 1:

$$\left|\frac{d^k\phi(x)}{dx^k}\right| \geq 1 \text{ for all } x \in [a,b]. \tag{2.35}$$

   We want an upper bound on $I(\lambda) = \int_a^b e^{i\lambda\phi(x)}dx$ that depends only on λ (not on a or b).

2. **Method:** We do the change of variables. With $x \to \lambda^{-1/k}x'$, and $dx \to \lambda^{-1/k}dx'$, we have:

$$I(\lambda) = \int_a^b e^{i\lambda\phi(x)}dx \implies I(\lambda) = \lambda^{-1/k}\int_{\lambda^{1/k}a}^{\lambda^{1/k}b} e^{i\lambda\phi(\lambda^{-1/k}x')}dx'. \tag{2.36}$$

   Now, we do Taylor expansion about $x' = 0$ for ϕ:

$$\phi(\lambda^{-1/k}x') = \phi(0) + \phi'(0)(\lambda^{-1/k}x') + \frac{1}{2!}\phi''(0)(\lambda^{-1/k}x')^2 + \cdots \frac{\phi^{(k)}(\xi)}{k!}(\lambda^{-1/k}x')^k. \tag{2.37}$$

   where ξ lies between 0 and $\lambda^{-1/k}x'$. Hence, we have:

$$\phi(\lambda^{-1/k}x') = \phi(0) + \sum_{j=1}^{k-1}\frac{\phi(j)(0)}{j!}\lambda^{-j/k}(x')^j + \frac{\phi^{(k)}(\xi)}{k!}\lambda^{-1}(x')^k. \tag{2.38}$$

   Substitute it back:

$$\begin{aligned} I(\lambda) &= \lambda^{-1/k}\int_{\lambda^{1/k}a}^{\lambda^{1/k}b} e^{i\lambda\phi(\lambda^{-1/k}x')}dx' \\ &= \lambda^{-1/k}\int_{\lambda^{1/k}a}^{\lambda^{1/k}b} \exp\left[i\lambda\left(\phi(0) + \sum_{j=1}^{k-1}\frac{\phi^{(j)}(0)}{j!}\lambda^{-j/k}(x')^j + \frac{\phi^{(k)}(\xi)}{k!}\lambda^{-1}(x')^k\right)\right]dx'. \\ &= \lambda^{-1/k}e^{i\lambda\phi(0)}\int_{\lambda^{1/k}a}^{\lambda^{1/k}b} \exp\left[i\left(\sum_{j=1}^{k-1}\frac{\phi^{(j)}(0)}{j!}\lambda^{1-j/k}(x')^j + \frac{\phi^{(k)}(\xi)}{k!}(x')^k\right)\right]dx'. \end{aligned} \tag{2.39}$$

Thus, we have:

$$I(\lambda) = \underbrace{\lambda^{-1/k}}_{\text{Scaling}} \underbrace{e^{i\lambda\phi(0)}}_{\text{Global phase}} \underbrace{\int_{\lambda^{1/k}a}^{\lambda^{1/k}b} e^{i\Psi_\lambda(x')}dx'}_{\text{Oscillatory integral}}. \tag{2.40}$$

where $\Psi_\lambda(x') = \sum_{j=1}^{k-1} c_j \lambda^{1-j/k}(x')^j + \frac{\phi^k(\xi)}{k!}(x')^k$ and $c_j = \frac{\phi^{(j)}(0)}{j!}$.

Now, to estimate $I(\lambda)$, we take the magnitude:

$$\begin{aligned} |I(\lambda)| &= \lambda^{-1/k} \cdot |e^{i\lambda\phi(0)}| \cdot \left| \int e^{i\Psi_\lambda(x')}dx' \right|. \\ &\leq C_k \lambda^{-1/k}. \end{aligned} \tag{2.41}$$

where phase factor magnitude $|e^{i\lambda\phi(0)}| = 1$ ($|e^{i\theta} \cdot z| = |z|, \forall\theta \in \mathbb{R}, z \in \mathbb{C}$), and the oscillatory integral can be bounded: $\left| \int e^{i\Psi_\lambda(x')}dx' \right| \leq C_k$.

[Proposition 2] (Van der Corput's Lemma) Let $\phi \in C^k(a,b)$ be real-valued, and assume $|\phi^{(k)}(x)| \geq 1$ for all $x \in (a,b)$. Then:

$$\left| \int_a^b e^{i\lambda\phi(x)}dx \right| \leq C_k \lambda^{-1/k}, \quad \forall\lambda > 0. \tag{2.42}$$

where C_k depends only on k, provided either:

(i) $k = 1$ and $\phi'(x)$ is monotonic, or

(ii) $k \geq 2$ (no additional assumptions).

The lemma quantifies how rapidly oscillatory integrals decay when the phase function $\phi(x)$ has controlled derivatives. The decay rate depends on which derivative first becomes non-vanishing. As it mentioned, we have two cases here.

Case i: $k = 1$, and $\phi'(x)$ is monotonic:

Proof. Assume ① : $\phi \in C^1([a,b])$, real-valued; ② : $|\phi'(x) \geq 1|, \forall x \in [a,b]$; ③ : $\phi'(x)$ is monotonic on $[a,b]$ (either non-decreasing or non-increasing).

1. *Step 1: Integration by Parts:* We express the oscillatory integral in terms of a differential operator:

$$\frac{d}{dx}\left(e^{i\lambda\phi(x)}\right) = i\lambda\phi'(x)e^{i\lambda\phi(x)} \implies \underbrace{e^{i\lambda\phi(x)}}_{\text{Df(x)}} = (i\lambda\phi'(x))^{-1} \underbrace{\frac{d}{dx}\left(e^{i\lambda\phi(x)}\right)}_{df/dx}. \tag{2.43}$$

This gives us:

$$Df(x) = (i\lambda\phi'(x))^{-1}df/dx \quad \text{and} \quad {}^tDf(x) = -d/dx(f(x)/i\lambda\phi'(x)). \tag{2.44}$$

Now, we do the integration by parts, with $f = e^{i\lambda\phi(x)}$:

$$\int_a^b e^{i\lambda\phi}dx = \int_a^b D(e^{i\lambda\phi})dx = \int_a^b (i\lambda\phi'(x))^{-1}df/dxdx.$$
$$= (i\lambda\phi'(x))^{-1}\cdot e^{i\lambda\phi(x)}\Big|_a^b - \int_a^b e^{i\lambda\phi(x)}\cdot\frac{d}{dx}(i\lambda\phi'(x))^{-1}dx. \quad (2.45)$$

Let $f \equiv 1$, for the constant function, we have:

$$Df(x) = (i\lambda\phi'(x))^{-1}df/dx \implies D(1) = (i\lambda\phi'(x))^{-1}. \quad (2.46)$$
$${}^tDf(x) = -d/dx(f/i\lambda\phi'(x)) \implies {}^tD(1) = -d/dx(i\lambda\phi'(x))^{-1}. \quad (2.47)$$

which leads to:

$$\int_a^b e^{i\lambda\phi(x)}dx = \int_a^b e^{i\lambda\phi(x)}\cdot{}^tD(f(x))dx + (i\lambda\phi'(x))^{-1}\cdot e^{i\lambda\phi(x)}\Big|_a^b.$$
$$= \underbrace{\int_a^b e^{i\lambda\phi(x)}\cdot{}^tD(1)dx}_{\text{Integral term}} + \underbrace{D(1)\cdot e^{i\lambda\phi(x)}\Big|_a^b}_{\text{Boundary term}}. \quad (2.48)$$

2. *Step 2: Boundary Term Estimate:* For the boundary term:

$$\left[e^{i\lambda\phi(x)}D(1)\right]_a^b = \left[e^{i\lambda\phi(x)}(i\lambda\phi'(x))^{-1}\right]_a^b = e^{i\lambda\phi(b)}(i\lambda\phi'(b))^{-1} - e^{i\lambda\phi(a)}(i\lambda\phi'(a))^{-1}. \quad (2.49)$$

Since $|\phi'(x)| \geq 1$, each of the term satisfies:

$$\left|e^{i\lambda\phi(x)}(i\lambda\phi'(x))^{-1}\right| \leq \frac{1}{\lambda|\phi'(x)|} \leq \frac{1}{\lambda}. \quad (2.50)$$

There're two points of the ends, a and b. This gives us:

$$\left[e^{i\lambda\phi(x)}D(1)\right]_a^b \leq \frac{1}{\lambda} + \frac{1}{\lambda} = \frac{2}{\lambda}. \quad (2.51)$$

3. *Step 3: Integral Term Estimate:* For the integration term:

$$\left|\int_a^b e^{i\lambda\phi(x)}\cdot{}^tD(1)dx\right| = \left|\int_a^b e^{i\lambda\phi(x)}\cdot(i\lambda)^{-1}\frac{d}{dx}\left(\frac{1}{\phi'(x)}\right)dx\right|.$$
$$\leq \lambda^{-1}\int_a^b\left|\frac{d}{dx}\left(\frac{1}{\phi'(x)}\right)\right|dx = \lambda^{-1}\left|\int_a^b\frac{d}{dx}\left(\frac{1}{\phi'(x)}\right)dx\right|. \quad (2.52)$$

Thus:

$$\left|\int_a^b e^{i\lambda\phi(x)}\cdot{}^tD(1)dx\right| \leq (\lambda)^{-1}\int_a^b\left|\frac{\phi''(x)}{(\phi'(x))^2}\right|dx. \quad (2.53)$$

As we remember, the assumption that ϕ' is monotonic (from Prop 2, ref: 2.2) implies ϕ'' does

not change sign on $[a, b]$. This lets us drop the absolute value inside the integral and evaluate it explicitly. We have two cases here:

i. **Case A:** $\phi''(x) \geq 0$ (e.g., ϕ' non-decreasing): Since $\phi''(x) \geq 0$, we have:

$$\left| \frac{\phi''(x)}{\left(\phi'(x)\right)^2} \right| = \frac{\phi''(x)}{\left(\phi'(x)\right)^2}. \tag{2.54}$$

The integral simplifies via the chain rule:

$$\int_a^b \frac{\phi''(x)}{(\phi'(x))^2} dx = \int_a^b \frac{d}{dx}\left(-\frac{1}{\phi'(x)}\right) dx = \left[-\frac{1}{\phi'(x)}\right]_a^b = \frac{1}{\phi'(a)} - \frac{1}{\phi'(b)}. \tag{2.55}$$

Since $|\phi'(x| \geq 1$, this gives us:

$$\left| \frac{1}{\phi'(a)} - \frac{1}{\phi'(b)} \right| \leq 1. \tag{2.56}$$

ii. **Case B:** $\phi''(x) \leq 0$ (e.g., ϕ' non-increasing): Now, we have:

$$\left| \frac{\phi''(x)}{\left(\phi'(x)\right)^2} \right| = -\frac{\phi''(x)}{\left(\phi'(x)\right)^2}. \tag{2.57}$$

The integral becomes:

$$\int_a^b -\frac{\phi''(x)}{(\phi'(x))^2} dx = \int_a^b \frac{d}{dx}\left(\frac{1}{\phi'(x)}\right) dx = \left[\frac{1}{\phi'(x)}\right]_a^b = \frac{1}{\phi'(b)} - \frac{1}{\phi'(a)}. \tag{2.58}$$

Since $|\phi'(x| \geq 1$, this also gives us:

$$\left| \frac{1}{\phi'(a)} - \frac{1}{\phi'(b)} \right| \leq 1. \tag{2.59}$$

Combine the two cases together, we thus have:

$$\left| \int_a^b e^{i\lambda\phi(x)} \cdot {}^t D(1) \right| \leq \frac{1}{\lambda}. \tag{2.60}$$

4. *Step 4: Final Bound:* Add up the two contributions 2.51 and 2.60, we have:

$$\left| \int_a^b e^{i\lambda\phi(x)} dx \right| \leq \underbrace{\frac{2}{\lambda}}_{\text{Boundary}} + \underbrace{\frac{1}{\lambda}}_{\text{Integral}} = \frac{3}{\lambda}. \tag{2.61}$$

which gives us $C_1 = 3$.

□

Case ii: $k \geq 2$ (no additional assumptions):

Key Idea: Here, we have the inductive proof of Prop 2 (ref: 2.2) for higher derivatives ($k \geq 2$). For oscillatory integrals where the first non-vanishing derivative is the k-th, we want to:

i. Split the integral around the point c where $|\phi^{(k)}|$ is minimized.

ii. Balance two competing terms:

- The contribution from the critical region (width δ).
- The contribution from non-critical regions (decay $\sim (\lambda\delta)^{-1/k}$).

Proof. Assume the lemma holds for some $k \geq 2$. Let $c \in [a, b]$ be a critical point where $|\phi^{(k)}(x)|$ is minimized. If $\phi^{(k)}(c) = 0$, then, outside the interval $(c - \delta, c + \delta)$, we have $|\phi^{(k)}(x)| \geq \delta$.

1. *Step 1: Identify Critical Point c:* Find $c \in [a, b]$ where $|\phi^{(k)}(x)|$ is minimized. Here, we have two cases:

   i. *Case 1:* $|\phi^{(k)}(c)| \geq \delta$. The k-th derivative is uniformly bounded below by δ on $[a, b]$. This means that $|\phi^{(k)}(x)| \geq \delta$ is everywhere. The result follows directly from the induction hypothesis. We can show this briefly:

   1.1 **Rescale:** From Prop 2 (ref: 2.2), we already assume that if $|\phi^{(k)}(x)| \geq 1$ everywhere, then the integral decays like $\lambda^{-1/k}$. Here we have $|\phi^{(k)}(x)| \geq \delta$, so we rescale. We define $\tilde{\phi}(x) = \phi(x)/\delta$. Now $|\tilde{\phi}^k(x)| \geq 1$, the inetgral becomes:

   $$\int_a^b e^{i\lambda\phi(x)} dx = \int_a^b e^{i(\lambda\delta)\tilde{\phi}(x)} dx. \tag{2.62}$$

   1.2 **Inductive hypothesis:** Here, we apply it to $\tilde{\phi}$ with $\tilde{\lambda} = \lambda\delta$:

   $$\left| \int_a^b e^{i(\lambda\delta)\tilde{\phi}(x)} dx \right| \leq C_k (\lambda\delta)^{-1/k}. \tag{2.63}$$

   Since δ is fixed constant in this case, this is $\mathcal{O}(\lambda^{-1/k})$.

   ii. *Case 2:* $|\phi^{(k)}(c)| < \delta$. The k-th derivative is small near c (a "near-critical" point), which is the "degenerate" scenario. We proceed to decompose the integral.

2. *Step 2: Split the Integral:* Followed by the case 2, we split the integral into three parts:

$$\int_a^b = \underbrace{\int_a^{c-\delta}}_{\text{Left}} + \underbrace{\int_{c-\delta}^{c+\delta}}_{\text{Central}} + \underbrace{\int_{c+\delta}^b}_{\text{Right}}. \tag{2.64}$$

3. *Step 3: Estimate Each Part:*

i. Central Interval Estimate ($|x-c| \le \delta$): So, inside $(c-\delta, c+\delta)$:

$$\left| \int_{c-\delta}^{c+\delta} e^{i\lambda\phi(x)} dx \right| \le \int_{c-\delta}^{c+\delta} 1 dx = 2\delta. \tag{2.65}$$

Note that there's no oscillation cancellation here, and this is a trivial bound.

ii. Left/Right integrals ($|x-c| \ge \delta$): For $x \in [a, c-\delta] \cup [c+\delta, b]$, we estimate the two outside pieces via induction, which is similiar as what we have for case 1. We start with:

$$\left| \int_a^b e^{i\lambda\phi(x)} dx \right| \le C_k \lambda^{-1/k}. \tag{2.66}$$

- On $[a, c-\delta]$, we have:

$$\left|\phi^{(k)}(x)\right| \ge \delta, \quad \left| \int_a^{c-\delta} e^{i\lambda\phi(x)} dx \right| \le C_k (\lambda\delta)^{-1/k}. \tag{2.67}$$

- On $[c+\delta, b]$, we have:

$$|\phi^{(k)}(x)| \le \delta, \quad \left| \int_{c+\delta}^b e^{i\lambda\phi(x)} dx \right| \le C_k (\lambda\delta)^{-1/k}. \tag{2.68}$$

4. *Step 4: Combined Bound:* Now, we combine the central interval, as well as the left and right parts:

$$\begin{aligned} \left| \int_a^b e^{i\lambda\phi(x)} dx \right| &= \underbrace{\left| \int_a^{c-\delta} e^{i\lambda\phi(x)} dx \right|}_{\text{Left}} + \underbrace{\left| \int_{c-\delta}^{c+\delta} e^{i\lambda\phi(x)} dx \right|}_{\text{Central}} + \underbrace{\left| \int_{c+d}^b e^{i\lambda\phi(x)} dx \right|}_{\text{Right}} \\ &\le \underbrace{2C_k(\lambda\delta)^{-1/k}}_{①} + \underbrace{2\delta}_{②}. \end{aligned} \tag{2.69}$$

Either ① or ② can be dominated:

i. if $\delta \uparrow$, ② dominates. This means that the critical region dominates, and the decay from non-critical regions is wasted.

ii. if $\delta \downarrow$, ① dominates. The non-critical term $2C_k(\lambda\delta)^{-1/k}$ blows up, even though the critical region shrinks.

We want to balance the two terms to ensures neither term dominates unnecessarily, giving the best possible decay rate. This can be shown in the next step.

5. *Step 5: Optimal δ Choice:* To minimize the total error, we balance the terms by setting them equal:

$$2C_k(\lambda\delta)^{-1/k} = 2\delta \implies C_k(\lambda\delta)^{-1/k} = \delta. \tag{2.70}$$

This gives:

$$C_k \lambda^{-1/k} = \delta^{1+1/k} \implies \delta = C_k^{k/(k+1)} \lambda^{-1/(k+1)}. \tag{2.71}$$

However, since C_k is a constant depending only on k, we can absorb $C_k^{k/(k+1)}$ into the final constant, giving the scaling law:

$$\delta \sim \lambda^{-1/(k+1)}. \tag{2.72}$$

Yielding:

$$\left| \int_a^b e^{i\lambda\phi(x)} dx \right| \leq (2C_k + 2)\lambda^{-1/(k+1)}. \tag{2.73}$$

6. *Step 6: Solving the Recurrence for C_k:* Given the recurrence relation from Step 5:

$$C_{k+1} = 2C_k + 2 \quad \text{with} \quad C_1 = 3, \tag{2.74}$$

we solve it explicitly to obtain a closed-form expression for C_k. We want to find the homogeneous solution by solving $C_{k+1}^{(h)} = 2C_k^{(h)}$:

$$C_k^{(h)} = A \cdot 2^{k-1} \quad (A \text{ is a constant}). \tag{2.75}$$

For a particular solution, we assume a constant solution $C_k^{(p)} = C$:

$$C = 2C + 2 \implies C = -2. \tag{2.76}$$

For the general solution, we combine homogeneous and particular solutions together:

$$C_k = A \cdot 2^{k-1} - 2. \tag{2.77}$$

To determine A, we use the base case $C_1 = 3$:

$$C_1 = A \cdot 2^0 - 2 = A - 2 = 3 \implies A = 5. \tag{2.78}$$

This gives us the final solution:

$$C_k = 5 \cdot 2^{k-1} - 2 \quad \text{for all} \quad k \geq 1. \tag{2.79}$$

□

We just finished the proof of Prop 2 (ref: 2.2). However, it does have limitation: it assumes $\psi \equiv 1$, whereas real-world problems require $\psi(x)$ to vary. Moreover, we want to understand on quantifing how $\psi(x)$ affects the decay rate. These are the reasons why we need to have the below Corollary.

Corollary 2.1 (Amplitude with Controlled Derivatives)**.** *Let $\phi \in C^k([a,b])$ satisfy $|\phi^{(k)}(x)| \geq 1$, and*

let $\psi \in C^1([a,b])$. Then:

$$\left| \int_a^b e^{i\lambda\phi(x)} \psi(x) dx \right| \leq C_k \lambda^{-1/k} \left(|\psi(b)| + \int_a^b |\psi'(x)| dx \right). \tag{2.80}$$

where C_k is the same constant as in Proposition 2.

Proof. Define the oscillatory integral function:

$$F(x) := \int_a^x e^{i\lambda\phi(t)} dt. \tag{2.81}$$

From Prop 2 (ref: 2.2), we have the uniform bound:

$$|F(x)| \leq C_k \lambda^{-1/k} \quad \forall x \in [a,b]. \tag{2.82}$$

Now integrate by parts:

$$\int_a^b e^{i\lambda\phi(x)} \psi(x) dx = \int_a^b F'(x)\psi(x) dx = [F(x)\psi(x)]_a^b - \int_a^b F(x)\psi'(x) dx. \tag{2.83}$$

Taking absolute values and applying the bound 2.82:

$$\begin{aligned} \left| \int_a^b e^{i\lambda\phi(x)} \psi(x) dx \right| &\leq |F(b)\psi(b)| - |F(a)\psi(a)| + \int_a^b |F(x)||\psi'(x)| dx. \\ &\leq C_k \lambda^{-1/k} |\psi(b)| + C_k \lambda^{-1/k} \int_a^b |\psi'(x)| dx. \\ &= C_k \lambda^{-1/k} \left(|\psi(b)| + \int_a^b |\psi'(x)| dx \right). \end{aligned} \tag{2.84}$$

Note that $F(a) = 0$ by definition, which eliminates the $\psi(a)$ term. The result follows. □

2.3 Asymptotics

Intuition: Oscillatory integrals model wave interference, where critical points (stationary phase points) dominate the behavior as $\lambda \to \infty$. The asymptotic expansion separates the integral into a leading-order term (determined by the critical point), and lower-order corrections (higher powers of λ^{-1}).

i. **Role of Critical Points:** The behavior of $I(\lambda)$ as $\lambda \to \infty$ is dominated by points x_0 where $\phi'(x_0) = 0$ (critical points).

   - **If ψ is away from critical points:** $I(\lambda)$ decays faster than any power of λ^{-1} (Prop 1:2.1).
   - **If ψ is localized near a critical point:** $I(\lambda)$ has asymptotic expansion in powers of $\lambda^{-1/k}$, where k is the order of the critical point.

ii. **Order of the Critical Point (k):** The smallest integer $k \geq 2$ for which $\phi^{(k)}(x_0) \neq 0$ determines the expansion:

   - **Non-degenerate case (k = 2)** [1]**:** $\phi''(x) \neq 0$ gives expansion in $\lambda^{-1/2}$.
   - **Degenerate case (k > 2)** [2]**:** Higher-order vanishing gives slower decay $\lambda^{-1/k}$.

Proposition 3 (Stationary Phase Asymptotics) Suppose $k \geq 2$, and ϕ has a critical point at x_0 of order k, i.e.,

$$\phi(x_0) = \phi'(x_0) = \cdots = \phi^{(k-1)}(x_0) = 0, \quad \phi^{(k)}(x_0) \neq 0. \tag{2.85}$$

If ψ is supported in a sufficiently small neighborhood of x_0, then:

$$I(\lambda) = \int e^{i\lambda\phi(x)}\psi(x)dx \sim \lambda^{-1/k} \sum_{j=0}^{\infty} a_j \lambda^{-j/k}. \tag{2.86}$$

where the asymptotic expansion holds in the sense that for all nonnegative integers N and r:

$$\left(\frac{d}{d\lambda}\right)^r \left[I(\lambda) - \lambda^{-1/k} \sum_{j=0}^{N} a_j \lambda^{-j/k} \right] = \mathcal{O}\left(\lambda^{-r-(N+1)/k}\right) \quad \text{as } \lambda \to \infty. \tag{2.87}$$

2.3.1 *Case i: $k = 2$ for $\phi(x) = x^2$*

We give the proof for $k = 2$ for $\phi(x) = x^2$ first. As mentioned in the book, there're three steps.

[1] A critical point x_0 is non-degenerate if the second derivative (Hessian for $n \geq 2$) of ϕ is nonzero at x_0 (i.e., $\phi''(x_0) \neq 0$). This implies the phase ϕ has quadratic curvature near x_0, leading to $I(\lambda) \sim \lambda^{-1/2}$.

[2] A critical point x_0 is degenerate if $\phi''(x_0) = 0$ and the first non-zero derivative is of order $k > 2$ (i.e., $\phi^{(k)}(x_0) \neq 0$):

$$\phi''(x_0) = \phi'''(x_0) = \cdots = \phi^{(k-1)}(x_0) = 0, \quad \phi^{(k)}(x_0) \neq 0.$$

The phase is latter near x_0, resulting in slower decay $I(\lambda) \sim \lambda^{-1/k}$.

Proof. 1. *Step 1: First Observation:*

First Observation (for $k = 2$, $\phi(x) = x^2$, and $\psi(x) = x^\ell e^{-x^2}$):

$$\int_{-\infty}^{\infty} e^{i\lambda x^2} \underbrace{x^\ell e^{-x^2}}_{\psi(x)} dx \sim \lambda^{-(\ell+1)/2} \sum_{j=0}^{\infty} c_j^{(\ell)} \lambda^{-j}. \tag{2.88}$$

for any nonnegative integer ℓ; when ℓ is odd, the integral vanishes.

1.1 **Setup and Change of Variables:** We compute the integral for $\psi(x) = x^\ell e^{-x^2}$ (Gaussian-weighted monomials):

$$I(\lambda) = \int_{-\infty}^{\infty} e^{i\lambda x^2} x^\ell e^{-x^2} dx = \int_{-\infty}^{\infty} e^{-(1-i\lambda)x^2} x^\ell dx. \tag{2.89}$$

Now, we do change of variables: let $z = \sqrt{1-i\lambda} \cdot x$, and $dz = \sqrt{1-i\lambda} \cdot dx$, where $\sqrt{1-i\lambda}$ is the principal branch (cut along the negative real axis) [3].

$$I(\lambda) = \int_{-\infty}^{\infty} e^{-z^2} \left(\frac{z}{\sqrt{1-i\lambda}} \right)^\ell \frac{dz}{\sqrt{1-i\lambda}} = \underbrace{(1-i\lambda)^{-(\ell+1)/2}}_{\text{Prefactor}} \underbrace{\int_{-\infty}^{\infty} e^{-z^2} z^\ell \, dz}_{\text{Gaussian moment}}. \tag{2.90}$$

1.2 **Asymptotic Expansion of the Prefactor:** We rewrite the prefactor:

$$\begin{aligned} (1-i\lambda)^{-(\ell+1)/2} &= \left(-i\lambda \left(1 + \frac{1}{i\lambda} \right) \right)^{-(\ell+1)/2} \\ &= \lambda^{-(\ell+1)/2} \cdot \underbrace{(-i)^{-(\ell+1)/2}}_{\text{Principal branch}} \cdot \underbrace{\left(1 + \frac{1}{i\lambda} \right)^{-(\ell+1)/2}}_{\text{Binomial expansion}}. \end{aligned} \tag{2.91}$$

i. **Principle branch:** $(-i)^{-(\ell+1)/2} = e^{i\pi(\ell+1)/4}$ (since $-i = e^{-i\pi/2}$).

ii. **Binomial expansion:** For $|\lambda| > 1$, we expand the term using binomial series [4], with $z = 1/i\lambda$, and $\alpha = -(\ell+1)/2$: $\left(1 + \frac{1}{i\lambda}\right)^{-(\ell+1)/2} = \sum_{j=0}^{\infty} \binom{-(\ell+1)/2}{j} \left(\frac{1}{i\lambda}\right)^j$.

This converges absolutely for $|\lambda| > 1$. Now substitute the series back into the prefactor:

$$(1-i\lambda)^{-(\ell+1)/2} = \lambda^{-(\ell+1)/2} \cdot (-i)^{-(\ell+1)/2} \cdot \sum_{j=0}^{\infty} \binom{-(l+1)/2}{j} \left(\frac{1}{i\lambda} \right)^j. \tag{2.92}$$

[3] The principal branch defines $\sqrt{1-i\lambda} = |1-i\lambda|^{1/2} e^{i \arg(1-i\lambda)/2}$ where $\arg(1-i\lambda) \in (-\pi, \pi]$. For $\lambda > 0$, this ensures analyticity and consistent power expansions.

[4] The generalized binomial series:

$$(1+z)^\alpha = \sum_{j=0}^{\infty} \binom{\alpha}{j} z^j, \text{ where } \binom{\alpha}{j} = \frac{\alpha(\alpha-1)\cdots(\alpha-j+1)}{j!}.$$

Let $c_j^{(\ell)} = (-i)^{-(\ell+1)/2} \cdot \binom{-(\ell+1)/2}{j} \cdot i^{-j}$, we have:

$$(1 - i\lambda)^{-(\ell+1)/2} = \lambda^{-(\ell+1)/2} \sum_{j=0}^{\infty} c_j^{(\ell)} \lambda^{-j}. \tag{2.93}$$

1.3 **Gaussian Moment Evaluation:** For the Gaussian moment, we have two cases:

i. **Case ℓ is odd:** The integral vanishes (anti-symmetric integrand).

ii. **Case ℓ is even:** The integral evaluates to:

$$\Gamma\left(\frac{\ell+1}{2}\right), \quad \text{where } \Gamma \text{ is the Gamma function .} \tag{2.94}$$

For integer n, $\Gamma(n + 1/2) = \frac{(2n)!}{4^n n!}\sqrt{\pi}$.

1.4 **Combine Results:** Substitute the expansion 2.93 back, the integral becomes:

$$\int_{-\infty}^{\infty} e^{i\lambda x^2} x^\ell e^{-x^2} dx = \underbrace{\left(\lambda^{-(\ell+1)/2} \sum_{j=0}^{\infty} c_j^{(\ell)} \lambda^{-j}\right)}_{\text{Prefactor}} \underbrace{\int_{-\infty}^{\infty} e^{-z^2} z^\ell dz}_{\text{Gaussian moment}}. \tag{2.95}$$

Moreover, for ℓ is even, the Gaussian integral is a constant, so we can absorb it into c_j^ℓ:

$$\int_{-\infty}^{\infty} e^{i\lambda x^2} x^\ell e^{-x^2} dx \sim \lambda^{-(\ell+1)/2} \sum_{j=0}^{\infty} c_j^{(\ell)} \lambda^{-j}. \tag{2.96}$$

1.5 **Analytic Justification:** We want to rigorously justify the asymptotic expansion of the prefactor $(1 - i\lambda)^{-(\ell+1)/2}$ and its combination with the Gaussian moment.

1.5.1 **Branch Cut Handling:** The term $(1 - i\lambda)^{-(\ell+1)/2}$ is defined via the principal branch of the complex power function:

$$1 - i\lambda = \sqrt{1 + \lambda^2} e^{i\theta}, \quad \theta = -\arctan(\lambda). \tag{2.97}$$

giving

$$(1 - i\lambda)^{-(\ell+1)/2} = (1 + \lambda^2)^{-(\ell+1)/4} e^{-i(\ell+1)\theta/2}. \tag{2.98}$$

The principal branch ensures consistency for $\lambda > 0$ and avoids discontinuities across the negative real axis.

1.5.2 **Taylor Expansion Validity:** Rewrite the prefactor as:

$$(1 - i\lambda)^{-(\ell+1)/2} = \lambda^{-(\ell+1)/2} (w - i)^{-(\ell+1)/2}, \quad w = \lambda^{-1}. \tag{2.99}$$

The function $(w - i)^{-(\ell+1)/2}$ is analytic for $|w| < 1$ (since i lies outside this disk),

admitting a Taylor series:

$$(w-i)^{-(\ell+1)/2} = \sum_{j=0}^{\infty} c_j^{(\ell)} w^j, \quad c_j^{(\ell)} = \frac{(-1)^j}{j!} \frac{d^j}{dw^j} (w-i)^{-(\ell+1)/2} \Big|_{w=0}. \tag{2.100}$$

Substituting $w = \lambda^{-1}$ yields the asymptotic expansion:

$$(1-i\lambda)^{-(\ell+1)/2} = \lambda^{-(\ell+1)/2} \sum_{j=0}^{\infty} c_j^{(\ell)} \lambda^{-j}. \tag{2.101}$$

1.5.3 **Combining with Gaussian Moment:** For even ℓ, the integral evaluates to:

$$\int_{-\infty}^{\infty} e^{-z^2} z^\ell \, dz = \Gamma\left(\frac{\ell+1}{2}\right). \tag{2.102}$$

Thus, the full asymptotic expansion is:

$$I(\lambda) = \lambda^{-(\ell+1)/2} \left(\sum_{j=0}^{\infty} c_j^{(\ell)} \lambda^{-j} \right) \Gamma\left(\frac{\ell+1}{2}\right). \tag{2.103}$$

For $\ell = 0$, this reproduces Proposition 3's leading term:

$$I(\lambda) \sim \sqrt{\pi} e^{i\pi/4} \lambda^{-1/2}. \tag{2.104}$$

2. *Step 2: Second Observation - Estimate for General Amplitudes:* Now, we want to prove the decay estimate for oscillatory integrals.

Second Observation (for $k = 2$, $\phi(x) = x^2$, and $\eta(x) \in C_c^\infty(\mathbb{R})$):

$$\left| \int_{-\infty}^{\infty} \underbrace{e^{i\lambda x^2}}_{①} \underbrace{x^\ell}_{②} \underbrace{\eta(x)}_{③} dx \right| \leq A\lambda^{-1/2-\ell/2}. \tag{2.105}$$

where ① : the phase term; ② :a monomial weight that adjusts the amplitude of $\eta(x)$ near the critical point $x = 0$, and ③ : $\eta(x) \in C_c^\infty(\mathbb{R})$ is the general smooth, compactly supported amplitude (not just Gaussians).

2.1 **Splitting the integral:**

2.1.1 **Cutoff Function:** We choose a cutoff function $\alpha(x) \in C_c^\infty(\mathbb{R})$ such that:

$$\alpha(x) = \begin{cases} 1 & \text{for } |x| \leq 1, \\ 0 & \text{for } |x| \geq 2. \end{cases} \tag{2.106}$$

with smoothly transitions in $1 < |x| < 2$. We also want to rescale to $\alpha(x/\varepsilon)$ to localize near $x = 0$, where ε is a scale parameter.

i. $\alpha(x/\varepsilon) = 1$ **for** $|x| \leq \varepsilon$**:** This ensures the "near" region captures $x = 0$, the critical point where $\phi'(x) = 0$.

ii. $\alpha(x/\varepsilon) = 0$ **for** $|x| \geq 2\varepsilon$**:** This excludes the "far" region where oscillations are rapid.

iii. $\varepsilon < |x| < 2\varepsilon$**:** The smoothly transitions between 1 and 0 avoids discontinuities.

2.1.2 **Decomposition:** For a small $\varepsilon > 0$ (to be chosen later), we write:

$$\int e^{i\lambda x^2} x^\ell \eta(x) dx = \underbrace{\int e^{i\lambda x^2} x^\ell \eta(x)\alpha(x/\varepsilon) dx}_{\text{Near 0}} + \underbrace{\int e^{i\lambda x^2} x^\ell \eta(x)[1-\alpha(x/\varepsilon)] dx}_{\text{Far from 0}}. \quad (2.107)$$

Note: We pick a small $\varepsilon > 0$ here is to isolate the critical point's contribution while ensuring the far-region integral decays rapidly due to oscillations.

2.2 **Estimating the "Near 0" part:** We want to bound:

$$\left| \int_{\text{Near 0}} e^{i\lambda x^2} x^\ell \eta(x)\alpha(x/\varepsilon) dx \right|. \quad (2.108)$$

where near 0, we have $|x| \leq 2\varepsilon$, since $\alpha(x/\varepsilon) = 0$ for $|x| \geq 2\varepsilon$.

2.2.1 **Simplify the Integrand:** We want to analyze each term here:

i. **Phase term:** $|e^{i\lambda x^2}| = 1$ (the oscillatory part has magnitude 1).

ii. **Amplitude:** $\eta(x)$ is bounded since $|\eta(x)| \leq C$ for $x \in \text{supp}(\eta)$, and $\alpha(x/\varepsilon) \leq 1$ since $0 \leq \alpha \leq 1$.

Thus, we have:

$$\left| e^{i\lambda x^2} x^\ell \eta(x) a(x/\varepsilon) \right| \leq C|x|^\ell. \quad (2.109)$$

2.2.2 **Bound the Integral:** We integrate over $|x| \leq 2\varepsilon$:

$$\left| \int_{\text{Near 0}} \right| \leq \int_{|x|\leq 2\varepsilon} C|x|^\ell dx. \quad (2.110)$$

For $\ell \geq 0$:

$$\int_{|x|\leq 2\varepsilon} |x|^\ell dx = 2\int_0^{2\varepsilon} x^\ell dx = 2\left[\frac{x^{\ell+1}}{\ell+1}\right]_0^{2\varepsilon} = \frac{2(2\varepsilon)^{\ell+1}}{\ell+1}. \quad (2.111)$$

Thus, we have the result as:

$$\left| \int_{\text{Near 0}} \right| \leq \frac{2^{\ell+1}C}{\ell+1}\varepsilon^{\ell+1} = C'\varepsilon^{\ell+1}. \quad (2.112)$$

where $C' = \frac{2^{\ell+1}C}{\ell+1}$.

2.3 **Estimating the "Far from 0" part:** We want to bound:

$$\int e^{i\lambda x^2} \underbrace{x^\ell \eta(x)[1-\alpha(x/\varepsilon)]}_{f(x)} dx. \tag{2.113}$$

for $|x| \geq \varepsilon$, where oscillations are rapid.

2.3.1 **Operator Definition and Properties:** We define the differential operator D as:

$$Df = -\frac{1}{i\lambda}\frac{d}{dx}\left(\frac{f(x)}{2x}\right). \tag{2.114}$$

where D is designed to "transfer" the oscillation from $e^{i\lambda x^2}$ to f via integration by parts:

$$\int e^{i\lambda x^2} f(x)dx = \int e^{i\lambda x^2}(Df)(x)dx + \text{ boundary terms.} \tag{2.115}$$

Here, f is an arbitrary smooth function with sufficient decay (e.g., a test function or amplitude $\psi(x)$). Formally, this follows from:

$$\int e^{i\lambda x^2} f(x)dx = \int \left(\frac{1}{2i\lambda x}\frac{d}{dx}e^{i\lambda x^2}\right) f(x)dx = -\int e^{i\lambda x^2}\frac{d}{dx}\left(\frac{f(x)}{2i\lambda x}\right)dx. \tag{2.116}$$

As we just learned how to estimate integrals of the form 2.115 for a general (smooth, compactly supported) function f. We can also gain knowledge on how D interacts with the pure oscillatory factor $e^{i\lambda x^2}$ itself.

$$D(e^{i\lambda x^2}) = -\frac{1}{i\lambda}\frac{d}{dx}\left(\frac{e^{i\lambda x^2}}{2x}\right) = -\frac{1}{i\lambda}\left(\frac{2i\lambda x e^{i\lambda x^2}}{2x} - \frac{e^{i\lambda x^2}}{2x^2}\right) = e^{i\lambda x^2}\left(1 + \frac{1}{2i\lambda x^2}\right). \tag{2.117}$$

Thus:

$$D(e^{i\lambda x^2}) = e^{i\lambda x^2}(1 + \mathcal{O}((\lambda x^2)^{-1})). \tag{2.118}$$

The leading term $e^{i\lambda x^2}$ shows that D preserves the oscillation, and the correction $\mathcal{O}((\lambda x^2)^{-1})$ becomes small when $\lambda x^2 \gg 1$.

2.3.2 **Iterated Application:** For $f(x) = x^\ell \eta(x)[1-\alpha(x/\varepsilon)]$, apply D N times:

$$\int e^{i\lambda x^2} f dx = \int e^{i\lambda x^2}(D^N f)dx. \tag{2.119}$$

The boundary terms vanish because $f = x^\ell \eta(x)[1-\alpha(x/\varepsilon)]$ is compactly supported (due to $\eta \in C_c^\infty$ and α vanishing for $|x| \geq 2\varepsilon$).

$$\int_{\text{Far}} e^{i\lambda x^2} f\, dx = \frac{1}{(i\lambda)^N}\int e^{i\lambda x^2}\frac{d^N}{dx^N}\left(\frac{f(x)}{(2x)^N}\right)dx. \tag{2.120}$$

which gives the explicit form:

$$D^N f = \frac{1}{(i\lambda)^N} \frac{d^N}{dx^N} \left(\frac{f(x)}{(2x)^N} \right). \tag{2.121}$$

2.3.3 **Derivative Control and Estimation:** Using the Leibniz rule, we have:

$$\frac{d^N}{dx^N} \left(\frac{f(x)}{x^N} \right) = \underbrace{\sum_{k=0}^{N} \binom{N}{k} \frac{d^{N-k}}{dx^{N-k}} [x^\ell \eta(x)]}_{①} \cdot \underbrace{\frac{d^k}{dx^k} [1 - \alpha(x/\varepsilon)]}_{②} \cdot \underbrace{x^{-N}}_{③}. \tag{2.122}$$

Now, we want to analyze (2.122) term by term:

i. ②: for $|x| \geq \varepsilon$, $1 - \alpha(x/\varepsilon)$ is smooth, where derivatives decay as $\varepsilon^{-k} = \lambda^{k/2}$.

ii. ① and ③: The worst-case term has $k = 0$ (no derivatives on α):

$$\left| \frac{d^N}{dx^N} \left(\frac{x^\ell \eta(x)}{x^N} \right) \right| \leq C_N |x|^{\ell - 2N}. \tag{2.123}$$

where $\frac{d^N}{dx^N}(x^{\ell-N}) \sim x^{\ell-2N}$, and derivatives of η are controlled. Moreover, the constant C_N depends on: $\|\eta^{(k)}\|_{L^\infty}$ for $0 \leq k \leq N$ and $\|\alpha^{(k)}\|_{L^\infty}$ for $1 \leq k \leq N$.

2.3.4 **Integral Convergence:** Now, we want to understand the convergence condition for (2.123), we have:

$$\int_{|x| \geq \varepsilon} |x|^{\ell - 2N} dx \text{ converges if } \ell - 2N < -1 \implies N > \frac{\ell + 1}{2}. \tag{2.124}$$

This then gives us the below result:

$$\int_{|x| \geq \varepsilon} |x|^{\ell - 2N} dx = 2 \int_{\varepsilon}^{\infty} x^{\ell - 2N} dx = \frac{2\varepsilon^{\ell - 2N + 1}}{2N - \ell - 1}. \tag{2.125}$$

So, we have the final estimate:

$$\left| \int_{\text{Far}} \right| \leq \frac{C_N}{\lambda^N} \cdot \frac{2\varepsilon^{\ell - 2N + 1}}{2N - \ell - 1} = C_N'' \lambda^{-N} \varepsilon^{\ell - 2N + 1}. \tag{2.126}$$

2.4 **Balancing Terms:** Now, we want to balance the terms in 2.107, with the estimates from 2.112, and 2.126. We choose $\varepsilon = \lambda^{-1/2}$.

$$\begin{aligned} \int e^{i\lambda x^2} x^\ell \eta(x) dx &= \underbrace{\int e^{i\lambda x^2} x^\ell \eta(x) \alpha(x/\varepsilon) dx}_{\text{Near 0}} + \underbrace{\int e^{i\lambda x^2} x^\ell \eta(x) [1 - \alpha(x/\varepsilon)] dx}_{\text{Far from 0}}. \\ &\leq C' \varepsilon^{\ell+1} + C_N'' \lambda^{-N} \varepsilon^{\ell - 2N + 1} = C' \lambda^{-(\ell+1)/2} + C_N'' \lambda^{-(\ell+1)/2}. \end{aligned} \tag{2.127}$$

Note: We choose $\varepsilon = \lambda^{-1/2}$ is to balance the two error terms achieve the same decay

rate. This then gives us:

$$\left| \int e^{i\lambda x^2} x^\ell \eta(x)\, dx \right| \leq (C' + C'')\lambda^{-(\ell+1)/2} = A\lambda^{-1/2-\ell/2}. \tag{2.128}$$

2.5 **Rapid Decay for Non-Critical Case:** For $g \in \mathcal{S}$ vanishing near $x = 0$:

$$\left| \int e^{i\lambda x^2} g(x)\, dx \right| \leq \frac{C_N}{\lambda^N} \int \left| \frac{d^N}{dx^N} \left(\frac{g(x)}{x^N} \right) \right| dx = \mathcal{O}(\lambda^{-N}). \tag{2.129}$$

We have no stationary phase here, because oscillations are rapid everywhere in $\text{supp}(g)$. Moreover, the inetgral is finite (since g is Schwartsz), so decay is $\mathcal{O}(\lambda^{-N})$ for any N.

3. *Step 3: General ψ via Taylor Expansion:* We analyze the oscillatory integral:

$$I(\lambda) = \int_{-\infty}^{\infty} e^{i\lambda x^2} \psi(x) dx. \tag{2.130}$$

where $\psi(x) \in C_c^\infty(\mathbb{R})$ is smooth and compactly supported near $x = 0$.

3.1 **Rewriting the Integral:** We introduce a damping factor e^{-x^2} and a smooth cutoff $\tilde{\psi}(x)$ (equal to 1 on $\text{supp}(\psi)$) to 2.131:

$$I(\lambda) = \int_{-\infty}^{\infty} \underbrace{e^{i\lambda x^2} e^{-x^2}}_{\text{Oscillatory + damping}} \cdot \underbrace{\left[e^{x^2} \psi(x) \right]}_{\text{Amplitude}} \cdot \underbrace{\tilde{\psi}(x)}_{\text{Cutoff}} dx. \tag{2.131}$$

where the damping e^{-x^2} ensures convenience for the Taylor expansion, and $\tilde{\psi}(x)$ restricts the domain to $\text{supp}(\psi)$ while preserving smoothness.

3.2 **Taylor Expansion of $e^{x^2}\psi(x)$:** Expand $e^{x^2}\psi(x)$ around $x = 0$ [5]:

$$e^{x^2}\psi(x) = \sum_{j=0}^{N} b_j x^j + x^{N+1} R_N(x) = P(x) + x^{N+1} R_N(x). \tag{2.132}$$

where $P(x)$ is simply the N-th order Taylor polynomial of the function $e^{x^2}\psi(x)$ centered at $x = 0$, with $b_j = \frac{1}{j!} \frac{d^j}{dx^j} \left(e^{x^2}\psi(x) \right) \Big|_{x=0}$, and $R_N(x)$ is smooth and bounded.

[5] For any smooth function $f(x) \in C^\infty(\mathbb{R})$, its finite Taylor expansion around $x = 0$ is:

$$f(x) = \sum_{j=-0}^{N} \frac{f^{(j)}(0)}{j!} x^j + R_N(x).$$

where $f^{(j)}(0) : j-$th derivative of f evaluated at 0, and $R_N(x)$, the remainder term, can be expressed as:

$$R_N(x) = \frac{f^{(N+1)}(\xi)}{(N+1)!} x^{N+1}, \text{ for some } \xi \text{ between 0 and } x.$$

3.3 **Substitution and Splitting the Integral:** Substitute 2.132 into 2.131:

$$\begin{aligned} I(\lambda) &= \int e^{i\lambda x^2} e^{-x^2} \left(P(x) + x^{N+1} R_N(x) \right) \tilde{\psi}(x) dx. \\ &= \underbrace{\int e^{i\lambda x^2} e^{-x^2} P(x) \tilde{\psi}(x) dx}_{\text{(A)}} + \underbrace{\int e^{i\lambda x^2} e^{-x^2} x^{N+1} R_N(x) \tilde{\psi}(x) dx}_{\text{(B)}}. \end{aligned} \tag{2.133}$$

Now, we focus on (A). We want to compare it to the same integral without $\tilde{\psi}(x)$:

$$(A) = \underbrace{\int e^{i\lambda x^2} e^{-x^2} P(x) dx}_{\text{Term (a)}} + \underbrace{\int e^{i\lambda x^2} e^{-x^2} P(x) \left(\tilde{\psi}(x) - 1 \right) dx}_{\text{Term (c)}}. \tag{2.134}$$

In this way, we can isolate the pure polynomial integral (Term (a)) and then have a cutoff correction (Term (c)). Now, combine with (B):

$$I(\lambda) = \underbrace{\sum_{j=0}^{N} b_j \int_{-\infty}^{\infty} e^{i\lambda x^2} e^{-x^2} x^j \, dx}_{\text{Term (a)}} + \underbrace{\int_{-\infty}^{\infty} e^{i\lambda x^2} e^{-x^2} x^{N+1} R_N(x) \tilde{\psi}(x) \, dx}_{\text{Term (b)}} + \underbrace{\int_{-\infty}^{\infty} e^{i\lambda x^2} e^{-x^2} P(x) \left[\tilde{\psi}(x) - 1 \right] \, dx}_{\text{Term (c)}}. \tag{2.135}$$

where:

i. **Term (a):** Captures the polynomial part of the expansion. Each monomial x^j contributes a $\lambda^{-(j+1)/2}$ term.

ii. **Term (b):** Handles the remainder $R_N(x)$. The factor x^{N+1} ensures higher-order decay.

iii. **Term (c):** Corrects for the cutoff $\tilde{\psi}$ outside $\text{supp}(\psi)$.Vanishes rapidly due to non-stationary phase.

3.4 **Analyzing Each Term:**

3.4.1 **Term (a): Polynomial Contributions:**
For each j, we want to compute:

$$I_j(\lambda) = \int_{-\infty}^{\infty} e^{i\lambda x^2} e^{-x^2} x^j dx = \int_{-\infty}^{\infty} e^{-(1-i\lambda)x^2} x^j dx. \tag{2.136}$$

We have different cases here:

i. **Case i:** $j = 0$ (No x^j term) [6]:

$$I_0(\lambda) = \int_{-\infty}^{\infty} e^{-(1-i\lambda)x^2} dx = \sqrt{\frac{\pi}{1-i\lambda}}. \tag{2.137}$$

This is valid when $\text{Re}(1 - i\lambda) > 0$, and its always true for $\lambda \in \mathbb{R}$. For the asymp-

[6] For $\text{Re}(\alpha) > 0$, the basic Gaussian integral is: $\int_{\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}$.

totics as $\lambda \to \infty$, we have:

$$I_0(\lambda) \sim \sqrt{\frac{\pi}{-i\lambda}} = \sqrt{\frac{\pi}{\lambda}} e^{-i\pi/4} = \mathcal{O}(\lambda^{-1/2}). \tag{2.138}$$

This gives the leading-order decay $\mathcal{O}(\lambda^{-1/2})$.

ii. **Case ii:** $j > 0$: (Higher-order terms):

- Odd j: By symmetry, $I_j(\lambda) = 0$ for odd j.
- Even $j = 2m$: We differentiate $I_0(\lambda)$ m times with respect to λ:

$$\frac{d^m}{d\lambda^m} I_0(\lambda) = (-i)^m \int_{-\infty}^{\infty} e^{-(1-i\lambda)x^2} x^{2m} dx. \tag{2.139}$$

Compute the m-th derivative of $I_0(\lambda) = \sqrt{\frac{\pi}{1-i\lambda}}$ with respect to λ:

$$\frac{d^m}{d\lambda^m} \sqrt{\frac{\pi}{1-i\lambda}} = \sqrt{\pi} \cdot \frac{(2m-1)!!}{2^m} \cdot \frac{i^m}{(1-i\lambda)^{m+1/2}}. \tag{2.140}$$

where $(2m-1)!! = (2m-1)(2m-3)\cdots 1$ is the double factorial. Thus:

$$I_{2m}(\lambda) = \sqrt{\pi} \cdot \frac{(2m-1)!!}{2^m} \cdot \underbrace{\left(\frac{i^m}{(-i)^m}\right)}_{(-1)^m} (1-i\lambda)^{-(m+1/2)}. \tag{2.141}$$

Now, we connect 2.141 to Gamma function, using $(2m-1)!! = \frac{2^m}{\sqrt{\pi}}\Gamma(m+1/2)$:

$$I_{2m}(\lambda) = \Gamma\left(m + \frac{1}{2}\right) \cdot (-1)^m (1-i\lambda)^{-(m+1/2)}. \tag{2.142}$$

For the asymptotics, we have the approximation as $\lambda \to \infty$:

$$(1-i\lambda)^{-(m+1/2)} \approx (-i\lambda)^{-(m+1/2)} = e^{i\pi(m+1/2)/2} \lambda^{-(m+1/2)}. \tag{2.143}$$

Substituting back:

$$I_{2m}(\lambda) \sim \frac{\Gamma(m+1/2)}{(-i\lambda)^{m+1/2}} = \mathcal{O}(\lambda^{-m-1/2}). \tag{2.144}$$

The decay is faster for larger m, consistent with $\lambda^{-1/2}$ for $m = 0$.

3.4.2 **Term (b): Remainder Estimate:** We have the below integral to analyze:

$$\text{Term (b)} = \int_{-\infty}^{\infty} \underbrace{e^{i\lambda x^2}}_{\text{Oscillatory}} \cdot \underbrace{e^{-x^2} x^{N+1} R_N(x)}_{\text{Amplitude } f(x)} dx. \tag{2.145}$$

where it has two competing factors:

i. Oscillatory Part: $e^{i\lambda x^2}$ oscillates rapidly when $\lambda x^2 \gg 1$ (e.g., $|x| \gg \lambda^{-1/2}$).

ii. Amplitude: $e^{-x^2} x^{N+1} R_N(x)$:
   - Behaves like x^{N+1} near $x = 0$, and the Taylor remainder $R_N(x) \sim \mathcal{O}(x^{N+2})$for smooth ψ.
   - Decays exponentially for $|x| \gg 1$.

The Integral balances when the oscillations $\sim$ amplitude. This means that:

i. For $|x| \ll \lambda^{-1/2}$: Oscillations are slow ($\lambda x^2 \ll 1$), the amplitude $\sim x^{N+1}$ is small.

ii. For $|x| \gg \lambda^{-1/2}$: Oscillations are rapid, but the amplitude decays exponentially.

The dominant region is $|x| \sim \lambda^{-1/2}$, where both effects are comparable. This leads us to do scaling analysis. We let $x = \lambda^{-1/2} y$. Then:

$$dx = \lambda^{-1/2} dy, \;\; x^{N+1} = \lambda^{-(N+1)/2} y^{N+1}, \;\; \lambda x^2 = y^2. \tag{2.146}$$

Substitute into Term (b):

$$\text{Term (b)} = \lambda^{-(N+1)/2} \int_{-\infty}^{\infty} e^{iy^2} \cdot e^{-\lambda^{-1} y^2} y^{N+1} R_N(\lambda^{-1/2} y) dy. \tag{2.147}$$

Now, we simplify for $\lambda \gg 1$. For the exponential term, we have $e^{-\lambda^{-1} y^2} \approx 1$ for $\lambda \gg 1$, since $\lambda^{-1} y^2 \ll 1$ in the dominant region $|y| \sim 1$. For the remainder term $R_N(\lambda^{-1/2} y)$ is smooth and bounded for $|y| \sim 1$. Thus, we have:

$$\text{Term (b)} = \lambda^{-(N+1)/2} \int_{-\infty}^{\infty} e^{iy^2} y^{N+1} R_N(\lambda^{-1/2} y) dy. \tag{2.148}$$

For the convergence of the integral, e^{iy^2} ensures convergence at $|y| \to \infty$ (via Riemann-Lebesgue lemma [7]). y^{N+1} grows polynomially, but $R_N(\lambda^{-1/2} y)$ is bounded. So the inetgral converges to a constant C_N (independent of λ). Thus:

$$\text{Term (b)} = \mathcal{O}(\lambda^{-(N+1)/2}). \tag{2.149}$$

To achieve an error $\mathcal{O}(\lambda^{-K})$, we can have:

$$\frac{N+1}{2} \geq K \implies N \geq 2K - 1. \tag{2.150}$$

3.4.2 **Term (c): Cutoff Function:** This term arises from the introduction of the smooth cutoff

[7] The Riemann-Lebesgue lemma states that if $f \in L^1(\mathbb{R})$ (e.g., $\int_{-\infty}^{\infty} |f(y)| dy < \infty$), then

$$\lim_{|\xi| \to \infty} \int_{-\infty}^{\infty} e^{i\xi y} f(y)\, dy = 0.$$

Rapid oscillations ($e^{i\xi y}$) cause the integral to vanish as $\xi \to \infty$.

function $\tilde{\psi}(x)$ and is given by:

$$\text{Term (c)} = \int_{-\infty}^{\infty} \underbrace{e^{i\lambda x^2}}_{\text{Oscillatory}} \cdot \underbrace{e^{-x^2} P(x)\,(\tilde{\psi}(x) - 1)}_{\text{Amplitude}}\, dx. \tag{2.151}$$

where $P(x) = \sum_{j=0}^{N} b_j x^j$ is the Taylor polynomial of $e^{x^2}\psi(x)$, and $\tilde{\psi}(x)$ is a smooth cutoff function with $\psi(\tilde{x}) = 1$ on $\text{supp}(\psi)$. The key properties are:

i. Oscillations: The phase $\phi(x) = \lambda x^2$ has no critical point ($\phi'(x) = 2\lambda x \neq 0$).

ii. Amplitude Behavior: The integrand $e^{-x^2} P(x)(\tilde{\psi}(x) - 1)$ is smooth and decays exponentially as $|x| \to \infty$.

iii. Non-Stationary Phase Lemma: By the lemma 2.1, such integrals decay faster than any power of λ:

$$|\text{Term (c)}| \leq C_k \lambda^{-k}, \text{ for any } k \geq 0. \tag{2.152}$$

iv. Superpolynomial Decay: The bound holds for all k, meaning the decay is faster than any polynomial power of λ. This is denoted as:

$$\text{Term (c)} = O(\lambda^{-\infty}). \tag{2.153}$$

3.5 **Final Asymptotic Expansion:**

3.5.1 **Decomposition of the integral:** We write $I(\lambda)$ as a sum of three terms:

$$I(\lambda) = \text{Term (a)} + \text{Term (b)} + \text{Term (c)}. \tag{2.154}$$

where:

i. **Term (a):** Polynomial contributions from the Taylor expansion of $e^{x^2}\psi(x)$.

ii. **Term (b):** Remainder term from the Taylor expansion.

iii. **Term (c):** Boundary term (decays faster than any polynomial in λ^{-1}).

3.5.2 **Asymptotic Expansion of Term (a):** The dominant contribution comes from Term (a), which involves even powers of x due to symmetry:

$$\text{Term (a)} = \sum_{\substack{j=0 \\ j \text{ even}}}^{\infty} b_j \int_{-\infty}^{\infty} e^{i\lambda x^2} e^{-x^2} x^j dx. \tag{2.155}$$

Each integral in the sum can be evaluated explicitly:

$$\int_{-\infty}^{\infty} e^{-(1-i\lambda)x^2} x^j dx = \Gamma\left(\frac{j+1}{2}\right)(1 - i\lambda)^{-(j+1)/2}. \tag{2.156}$$

For $\lambda \to \infty$, we expand $(1 - i\lambda)^{-(j+1)/2}$asymptotically, useing the generalized binomial

theorem[8]:

$$(1-i\lambda)^{-(j+1)/2} = \left[-i\lambda\left(1+\frac{1}{i\lambda}\right)\right]^{-(j+1)/2}$$
$$= (-i\lambda)^{-(j+1)/2}\left(1+\frac{1}{i\lambda}\right)^{-(j+1)/2}.$$
$$\sim (-i\lambda)^{-(j+1)/2}\left(1-\frac{j+1}{2}\cdot\frac{1}{i\lambda}+\frac{\left(-\frac{j+1}{2}\right)\left(-\frac{j+1}{2}-1\right)}{2!}\left(\frac{1}{i\lambda}\right)^2+\cdots\right). \quad (2.157)$$

which gives us:

$$(1-i\lambda)^{-(j+1)/2} \sim (-i\lambda)^{-(j+1)/2}\left(1-\frac{j+1}{2i\lambda}+\cdots\right). \quad (2.158)$$

Thus, the leading-order behavior of Term (a) is:

$$\text{Term (a)} \sim \sum_{\substack{j=0 \\ j \text{ even}}}^{\infty} b_j \Gamma\left(\frac{j+1}{2}\right)(-i\lambda)^{-(j+1)/2}. \quad (2.159)$$

Substituting back, the asymptotic expansion becomes:

$$\text{Term (a)} \sim \sum_{m=0}^{\infty} b_{2m}\Gamma\left(m+\frac{1}{2}\right)e^{-i\pi(2m+1)/4}\lambda^{-(m+1/2)}. \quad (2.160)$$

3.5.3 **Phase and Amplitude Simplification:** Let $j = 2m$ (even integers only). Then:

$$(-i\lambda)^{-(j+1)/2} = e^{-i\pi(j+1)/4}\lambda^{-(j+1)/2}. \quad (2.161)$$

Substituting back, the asymptotic expansion becomes:

$$\text{Term (a)} \sim \sum_{m=0}^{\infty} b_{2m}\Gamma\left(m+\frac{1}{2}\right)e^{-i\pi(2m+1)/4}\lambda^{-(m+1/2)}. \quad (2.162)$$

3.5.4 **Leading-Order Term ($m = 0$):** For $m = 0$ ($j = 0$), we recover the leading-order decay:

$$a_0 = b_0\Gamma\left(\frac{1}{2}\right)e^{-i\pi/4} = \psi(0)\sqrt{\pi}e^{-i\pi/4}. \quad (2.163)$$

This matches the stationary phase result for $\phi(x) = x^2$:

$$a_0 = \psi(0)\sqrt{\frac{2\pi}{-i\phi''(0)}} = \psi(0)\sqrt{\frac{\pi}{i}}. \quad (2.164)$$

[8]For $|z| < 1$ (here $z = \frac{1}{i\lambda}$), the expansion of $(1+z)^\alpha$ is: $(1+z)^\alpha \sim 1 + \alpha z + \frac{\alpha(\alpha-1)}{2!}z^2 + \cdots$

3.5.5 **Higher-Order Terms ($m \geq 1$):** For $m \geq 1$, the coefficiens are:

$$a_{2m} = b_{2m}\Gamma\left(m + \frac{1}{2}\right) e^{-i\pi(2m+1)/4}. \tag{2.165}$$

These terms contribute corrections of order $\mathcal{O}(\lambda^{-(m+1)/2})$.

3.5.6 **Subdominant Terms:**

i. **Term (b):** The remainder term decays as $\mathcal{O}(\lambda^{-(N+1)/2})$ for any truncation order N.

ii. **Term (c):** The boundary term is $\mathcal{O}(\lambda^{-\infty})$ (exponentially small).

3.5.7 **Final Asymptotic Series:** Combining all contributions, the full asymptotic expansion is:

$$I(\lambda) \sim \sum_{m=0}^{\infty} a_{2m}\lambda^{-(m+1/2)}, \;\; a_{2m} = b_{2m}\Gamma\left(m + \frac{1}{2}\right) e^{-i\pi(2m+1)/4}. \tag{2.166}$$

We let $j = 2m$ (since only even terms survive). Then $\lambda^{-(m+1/2)} = \lambda^{-1/2} \cdot \lambda^{-m}$, which matches the form in Prop 3, ref: 2.3 with $j = 2m$.

□

2.3.2 Case ii: General Case for $k = 2$ (Nondegenerate Critical Points)

Now, we want to extend the asymptotic expansion from the model case $\phi(x) = x^2$ to general phases ϕ with a nondegenerate critical point at x_0 (e.g., $\phi'(x_0) = 0$, $\phi''(x_0) \neq 0$).

Proof. 4. *Step 4: General Case for $k = 2$, Nondegenerate Critical Points:*

4.1 **Local Approximation Near x_0:**

4.1.1 **Taylor Expansion:** Near x_0, expand ϕ as:

$$\phi(x) = \phi(x_0) + \underbrace{\frac{\phi''(x_0)}{2}(x - x_0)^2}_{\text{Quadratic term}} + \mathcal{O}(|x - x_0|^3). \tag{2.167}$$

For simplicity, we assume $\phi(x_0) = 0$ (otherwise, factor out $e^{i\lambda\phi(x_0)}$).

4.1.2 **Rescaling:** Let $c = \phi''(x_0)/2 \neq 0$. Then:

$$\phi(x) = c(x - x_0)^2 \left[1 + \underbrace{\epsilon(x)}_{\mathcal{O}(|x-x_0|)}\right], \quad \text{where } |\epsilon(x)| < 1 \text{ near } x_0. \tag{2.168}$$

For $|x - x_0|$ small, $\varepsilon(x)$ is negligible, and $\phi(x) \approx c(x - x_0)^2$.

4.2 **Change of Variables:** We now want to transform $\phi(x)$ into a quadratic form cy^2 for some y. We start with the Taylor expansion 2.168:

$$\phi(x) = c(x - x_0)^2(1 + \varepsilon(x)), \quad c = \phi''(x_0)/2. \tag{2.169}$$

where $\varepsilon(x) = \mathcal{O}(|x - x_0|)$ captures the higher-order terms. Now, with $\phi(x) = cy^2$ for some y. Solve for y:

$$cy^2 = c(x - x_0)^2(1 + \varepsilon(x)) \implies y = (x - x_0)\sqrt{1 + \varepsilon(x)}. \tag{2.170}$$

In this way, the y absorbs the higher-order term $\varepsilon(x)$ into the new variable y. Moreover, the map $x \mapsto y$ is invertible near x_0, since $\sqrt{1 + \varepsilon(x)}$ is smooth and ≈ 1. This transform the integral:

$$\int e^{i\lambda\phi(x)}\psi(x)\,dx = \int e^{i\lambda cy^2}\tilde{\psi}(y)dy. \tag{2.171}$$

where $\tilde{\psi}(y) = \psi(x(y)) \cdot \left|\frac{dx}{dy}\right|$ is the transformed amplitude, and is compactly supported near the point $y = 0$. We then account for the change of variables. Start with the y expression in 2.171, and differentiate y w.r.t. x:

$$\left|\frac{dx}{dy}\right| = \left|\frac{dy}{dx}\right|^{-1} = \left|\sqrt{1 + \varepsilon(x)} + \frac{x - x_0}{2\sqrt{1 + \varepsilon(x)}}\varepsilon'(x)\right|^{-1}. \tag{2.172}$$

Near x_0, this simplifies to ≈ 1, since $\varepsilon(x)$ is small.

4.3 **Reduction to Model Case:**

Now, we want to apply for the known asymptotics for $\phi(y) = cy^2$.

4.3.1 **Apply Prop 3 (ref: 2.3):** The transformed integral matches the model case $\phi(y) = cy^2$:

$$\int e^{i\lambda cy^2}\tilde{\psi}(y)\,dy \sim \lambda^{-1/2}\sum_{j=0}^{\infty} a_j\lambda^{-j}. \tag{2.173}$$

with the leading term:

$$a_0 = \tilde{\psi}(0)\sqrt{\frac{2\pi}{-ic}} = \psi(x_0)\sqrt{\frac{2\pi}{-i\phi''(x_0)}}. \tag{2.174}$$

where the odd terms vanish if $\tilde{\psi}$ is even (by symmetry).

□

2.3.3 *Case iii: General Case for higher k*

5. *Step 5: Extension to Higher-Order Critical Points ($k \geq 3$)* Extend the asymptotic expansion to phases ϕ with degenerate critical points, where:

$$\phi'(x_0) = \cdots = \phi^{(k-1)}(x_0) = 0, \quad \phi^{(k)}(x_0) \neq 0 \quad (k \geq 3). \tag{2.175}$$

Note: Here, we have slower decay rate due to flatter curvature ($\lambda^{-1/k}$ vs. $\lambda^{-1/2}$), and we don't use Morse lemma here. We do explicit reduction to y^k. Moreover, we have odd and

even effects based on k.

5.1 **Local Approximation:** The Taylor series of $\phi(x)$ around x_0 is:

$$\phi(x) = \phi(x_0) + \sum_{m=k}^{\infty} \frac{\phi^{(m)(x_0)}}{m!}(x - x_0)^m. \tag{2.176}$$

Since we assume $\phi(x_0) = 0$, we write:

$$\phi(x) = \frac{\phi^{(k)}(x_0)}{k!}(x - x_0)^k \left[1 + \epsilon(x)\right], \quad \epsilon(x) = O(|x - x_0|). \tag{2.177}$$

where we assume $|\epsilon(x)| < 1$ in a neighborhood U, and we also have the dominant term as $\frac{\phi^{(k)}(x_0)}{k!}y^k$ after transformation.

5.2 **Change of Variables:** Define $y = (x - x_0)\left[1 + \epsilon(x)\right]^{1/k}$, yielding:

$$\phi(x) \mapsto \frac{\phi^{(k)}(x_0)}{k!}y^k. \tag{2.178}$$

Like in case ii, we have $(1 + \epsilon(x))^{1/k}$ is smooth and invertible near x_0 as well. This gives us the below transformed integral:

$$\int e^{i\lambda\phi(x)}\psi(x)dx = \int e^{i\lambda c y^k}\tilde{\psi}(y)dy, \quad c = \frac{\phi^{(k)}(x_0)}{k!}. \tag{2.179}$$

5.3 **Integral Asymptotics:** Using the identity, as indicated in the book:

$$\int_0^{\infty} e^{i\lambda c y^k} e^{-y^k} y^{\ell} dy = c_{k,\ell}(1 - i\lambda c)^{-(\ell+1)/k}. \tag{2.180}$$

We then obtain:

$$\int e^{i\lambda c y^k}\tilde{\psi}(y)dy \sim \lambda^{-1/k}\sum_{j=0}^{\infty} a_j \lambda^{-j/k}. \tag{2.181}$$

With the leading term:

$$a_0 = \psi(x_0) \cdot \frac{\Gamma(1/k)}{k}(-ic)^{-1/k}. \tag{2.182}$$

where the odd terms vanish if k is even and ψ is symmetric.

2.3.4 Asymptotic Expansion Remarks

Recall from Prop 3, ref 2.3, we have:

$$I(\lambda) = \int e^{i\lambda\phi(x)}\psi(x)dx \sim \lambda^{-1/k}\sum_{j=0}^{\infty} a_j \lambda^{-j/k}. \tag{2.183}$$

In this remark, we can have the below summary regarding 2.183.

i. **Coefficient Structure:**

- Each a_j depends on finitely many derivatives of ϕ and ψ at x_0.
- For $k = 2$, the leading term is explicit:

$$a_0 = \psi(x_0) \left(\frac{2\pi}{-i\phi''(x_0)} \right)^{1/2}. \tag{2.184}$$

- Higher coefficients a_j involve derivatives up to order $2j$.

ii. **Error Analysis:**

- Error bounds depend on:
  ① Derivatives of ϕ, ψ in $\text{supp}(\psi)$.
  ② Size of $\text{supp}(\psi)$.
  ③ Lower bound for $|\phi^{(k)}(x_0)|$.
- Sharpness: Achieved when $\phi(x) = x^k$ and ψ is Gaussian.

iii. **Physical Interpretation ($k = 2$):**

$$\int e^{i\phi(x)} \psi(x) dx \approx \underbrace{e^{i\phi(x_0)}}_{\text{Phase}} \cdot \underbrace{\left(\frac{2\pi}{-i\phi''(x_0)} \right)^{1/2}}_{\text{Curvature}} \cdot \underbrace{\psi(x_0)}_{\text{Amplitude}}. \tag{2.185}$$

where the phase is determined by $\phi(x_0)$, and its amplitude is weighted at x_0. Moreover, it also has a decay rate of $\sim |\phi''(x_0)|^{-1/2}$.

iv. **Symmetry Effects:**

- For even k: All odd coefficients vanish ($a_j = 0$ when j odd).
- Due to antisymmetry in $y \mapsto -y$ for y^k.

3 OSCILLATORY INTEGRALS OF THE FIRST KIND, SEVERAL VARIABLES

3.1 Localization

Intuition: In higher dimensions, the integral:

$$I(\lambda) = \int_{\mathbb{R}^n} e^{i\lambda\phi(\mathbf{x})}\psi(\mathbf{x})d\mathbf{x}. \tag{3.1}$$

decays rapidly $\mathcal{O}(\lambda^{-N})$ for any $N \geq 0$, since the rapid oscillations of $e^{i\lambda\phi(\mathbf{x})}$ cause cancellation through destructive interference. However, there're two exceptions:

i. **Critical points:** where $\nabla\phi(\mathbf{x}_0) = \mathbf{0}$, causing the phase to vary slowly. Formally, $\mathbf{x}_0$ is a critical point if:

$$(\nabla\phi)(\mathbf{x}_0) = \left(\frac{\partial\phi}{\partial x_1}, \ldots, \frac{\partial\phi}{\partial x_n}\right)\Bigg|_{\mathbf{x}=\mathbf{x}_0} = \mathbf{0}. \tag{3.2}$$

Near these points, oscillations are "stationary" and contribute dominantly to the integral.

ii. **Boundary contributions:** where ψ doesn't vanish at the domain boundary. These are typically avoided by assuming $\psi \in C_c^\infty(\mathbb{R}^n)$.

[Proposition 4: Non-Stationary Phase Decay in $\mathbb{R}^n$] Let:

i. $\phi \in C^\infty(\mathbb{R}^n)$ with $\nabla\phi(\mathbf{x}) \neq \mathbf{0}$ for all $\mathbf{x} \in \text{supp}(\psi)$,

ii. $\psi \in C_c^\infty(\mathbb{R}^n)$ (compactly supported).

Then:

$$I(\lambda) := \int_{\mathbb{R}^n} e^{i\lambda\phi(\mathbf{x})}\psi(\mathbf{x})\, d\mathbf{x} = \mathcal{O}(\lambda^{-N}), \quad \forall N \geq 0 \quad \text{as } \lambda \to \infty. \tag{3.3}$$

Proof.

1. *Step 1: Construct a Directional Cover:* For each $\mathbf{x}_0 \in \text{supp}(\psi)$, since $\nabla\phi(\mathbf{x}_0) \neq \mathbf{0}$, there exists a direction $\xi_{\mathbf{x}_0} \in S^{n-1}$ [9] such that:

$$\xi_{\mathbf{x}_0} \cdot \nabla\phi(\mathbf{x}_0) = |\nabla\phi(\mathbf{x}_0)| > 0. \tag{3.4}$$

By the continuity of $\nabla\phi$ and the fact that $\xi_{\mathbf{x}_0} \cdot \nabla\phi(\mathbf{x}_0) = |\nabla\phi(\mathbf{x}_0)| > 0$, there exists an open neighborhood $U_{\mathbf{x}_0}$ of $\mathbf{x}_0$ and a constant $c(\mathbf{x}_0) > 0$ such that:

$$\xi_{\mathbf{x}_0} \cdot \nabla\phi(\mathbf{x}) \geq c(\mathbf{x}_0) \quad \text{for all } \mathbf{x} \in U_{\mathbf{x}_0}. \tag{3.5}$$

[9] The notation S^{n-1} denotes the $(n-1)$-dimensional unit sphere in $\mathbb{R}^n$, defined by $\{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = 1\}$. Its dimension is $n-1$ because the constraint $\|\mathbf{x}\| = 1$ reduces the degrees of freedom by 1.

In particular, this implies:

$$\inf_{\mathbf{x}\in U_{\mathbf{x}_0}} |\xi_{\mathbf{x}_0} \cdot \nabla\phi(\mathbf{x})| \geq c(\mathbf{x}_0) > 0. \tag{3.6}$$

2. *Step 2: Build a Partition of Unity:* For each U_j, we construct $\eta_j \in C_c^\infty(U_j)$ with $0 \leq \eta_j \leq 1$ and $\sum_{j=1}^M \eta_j(\mathbf{x}) = 1$ on $\text{supp}(\psi)$. This allows us to do localization:

$$I(\lambda) = \sum_{j=1}^{M} \int_{U_j} e^{i\lambda\phi(\mathbf{x})}\psi(\mathbf{x})\eta_j(\mathbf{x})\, d\mathbf{x}. \tag{3.7}$$

where each integrand $\psi\eta_j$ is supported in U_j, and η_j is a smooth cutoff function. Moreover, the smoothness of η_j preserves the regularity of ψ, ensuring the subsequent phase estimates apply.

3. *Step 3: Coordinate Alignment and 1D Reduction:* For each j, perform the following steps:

   3.1 **Rotation to align ξ_j with e_1:** Let R_j be a rotation matrix mapping the direction ξ_j (from Step 1) to the standard basis vector $e_1 = (1, 0, \ldots, 0)$ in $\mathbb{R}^n$. Define the rotated coordinates $\mathbf{y} = R_j\mathbf{x}$, and the rotation R_j transforms the coordinates so that:

$$\xi_j(\text{original direction}) \xrightarrow{R_j} e_1(\text{new } y_1\text{-axis}) \tag{3.8}$$

   In these coordinates, the directional derivative becomes:

$$\partial_{y_1}\phi(R_j^{-1}\mathbf{y}) = \xi_j \cdot \nabla\phi(\mathbf{x}) \geq c_j > 0. \tag{3.9}$$

   where the inequality holds because $\xi_j \cdot \nabla\phi \geq c_j$ on U_j (*Step 1*). This is the key inequality needed to apply the 1D non-stationary phase estimate.

   3.2 **Change of variables:** Rewrite the localized integral in rotated coordinates [10]:

$$\int_{U_j} e^{i\lambda\phi(\mathbf{x})}\psi(\mathbf{x})\eta_j(\mathbf{x})\, d\mathbf{x} = \int e^{i\lambda\phi(R_j^{-1}\mathbf{y})} \underbrace{\psi(R_j^{-1}\mathbf{y})\eta_j(R_j^{-1}\mathbf{y})|\det R_j^{-1}|}_{\tilde{\psi}_j(\mathbf{y})}\, d\mathbf{y}. \tag{3.10}$$

   Note: $|\det R_j^{-1}| = 1$ (rotations preserve volume), and $\tilde{\psi}_j$ is smooth and compactly supported.

   3.3 **Iterated integration via Fubini:** Split $\mathbf{y} = (y_1, \mathbf{y}')$ where $\mathbf{y}' = (y_2, \ldots, y_n)$, and apply Fubini's theorem [11] with iterated integrals in any order:

$$\int_{\mathbb{R}^{n-1}} \left(\int_{-\infty}^{\infty} e^{i\lambda\phi(R_j^{-1}(y_1,\mathbf{y}'))}\tilde{\psi}_j(y_1, \mathbf{y}')\, dy_1 \right) d\mathbf{y}'. \tag{3.11}$$

[10] For any diffeomorphism $y = T(x)$, the integral transforms as: $\int f(x)dx = \int f(T^{-1}(y))\left|\det DT^{-1}(y)\right| dy$, where DT^{-1} is the Jacobian matrix of T^{-1}. Since rotations are orthogonal, $\left|\det R_j^{-1}\right| = 1$.

[11] Let $f(x) = f(x_1, \ldots, x_n)$ be a measurable function on $\mathbb{R}^n$. If either: f is non-negative, or f is absolutely integrable

For each fixed $\mathbf{y}'$, the inner integral (in y_1) is a 1D oscillatory integral:

$$I_{\mathbf{y}'}(\lambda) = \int e^{i\lambda\phi_{y'}(y_1)}\tilde{\psi}_j(y_1, \mathbf{y}')\, dy_1, \quad \text{where } \phi_{y'}(y_1) := \phi(R_j^{-1}(y_1, \mathbf{y}')). \tag{3.12}$$

By the rotation alignment, $|\partial_{y_1}\phi_{y'}(y_1)| = |\xi_j \cdot \nabla\phi| \geq c_j > 0$ for all $(y_1, \mathbf{y}')$ in the support of $\tilde{\psi}_j$.

3.4 **1D non-stationary phase estimate:** Apply **Prop 1 (1D decay), Ref:** (2.1) to $I_{\mathbf{y}'}(\lambda)$: For any $N \geq 0$, there exists $C_N(c_j) > 0$ such that:

$$|I_{\mathbf{y}'}(\lambda)| \leq C_N(c_j)\lambda^{-N} \sum_{k=0}^{N} \sup_{y_1} \left|\partial_{y_1}^k \tilde{\psi}_j(y_1, \mathbf{y}')\right|. \tag{3.13}$$

The bound is uniform in $\mathbf{y}'$ because $\tilde{\psi}_j$ is C^N with compact support.

3.5 **Integrate over $\mathbf{y}'$:** Since $\tilde{\psi}_j$ has compact support, integrate the 1D estimate over $\mathbf{y}'$ to conclude:

$$\left|\int_{U_j} e^{i\lambda\phi}\psi\eta_j\, d\mathbf{x}\right| \leq C_{j,N}\lambda^{-N}. \tag{3.14}$$

where $C_{j,N}$ depends on c_j, $\|\tilde{\psi}_j\|_{C^N}$, and the volume of $\mathrm{supp}(\tilde{\psi}_j)$.

4. *Step 4: Global Estimate:* Summing over j gives the final decay:

$$|I(\lambda)| \leq \sum_{j=1}^{M} C_{j,N}\lambda^{-N} \leq C_N\lambda^{-N}. \tag{3.15}$$

The constant C_N depends on the minimal c_j (measuring "non-stationarity"), the C^N norms of ψ and η_j, and the Jacobians $|\det R_j^{-1}|$. □

(e.g., $\int |f| < \infty$), then the integral over $\mathbb{R}^n$ can be computed as an iterated integral in any order:

$$\int_{\mathbb{R}^n} f(x)\, dx = \int_{-\infty}^{\infty} \left(\cdots \left(\int_{-\infty}^{\infty} f(x_1, \ldots, x_n)\, dx_1 \right) \cdots \right) dx_n.$$

3.2 Scaling

3.2.1 Multi-Dimensional Scaling

Intuition: The decay rate of oscillatory integrals in higher dimensions is governed by the most "balanced" oscillations, where no single direction dominates. The scaling $\lambda^{-1/k}$ emerges from the highest-order non-vanishing derivative of the phase function ϕ across all possible directions.

[Proposition 5: Multi-Dimensional van der Corput] Suppose for some multi-index α with $|\alpha| = k$:

$$|\partial^\alpha \phi(\mathbf{x})| \geq 1 \quad \forall \mathbf{x} \in \text{supp}(\psi). \tag{3.16}$$

Then:

$$\left| \int_{\mathbb{R}^n} e^{i\lambda\phi(\mathbf{x})} \psi(\mathbf{x})\, d\mathbf{x} \right| \leq C_k \lambda^{-1/k} (\|\psi\|_{L^\infty} + \|\nabla\psi\|_{L^1}). \tag{3.17}$$

Proof. We proceed through three main steps, mirroring the 1D case but incorporating directional derivatives:

1. *Step 1: Representation via Directional Derivatives:*

   1.1 **Homogeneous Polynomial Decomposition**: Any homogeneous polynomial $P(\mathbf{x})$ of degree k can be expressed through directional derivatives:

   $$P(\mathbf{x}) = \sum_{|\alpha|=k} c_\alpha \mathbf{x}^\alpha = \sum_{j=1}^{d(k,n)} a_j (\xi^j \cdot \mathbf{x})^k. \tag{3.18}$$

   i. **Standard basis (left):** Monomials $\mathbf{x}^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$, summed over all multi-indices with $|\alpha| = k$.

   ii. **Directional basis (right):** Terms $(\xi^j \cdot \mathbf{x})^k$, where each $\xi^j \in S^{n-1}$ is a fixed unit vector. Each term is the k-th power of a linear function. The number of terms needed is $d(k,n) = \binom{n+k-1}{k}$, the dimension of the space.

   1.2 **Directional Derivative Lower Bound**: At each $\mathbf{x}_0 \in \text{supp}(\psi)$, the condition $|\partial^\alpha \phi(\mathbf{x}_0)| \geq 1$ implies there exists $\xi \in S^{n-1}$ such that:

   $$|(\xi \cdot \nabla)^k \phi(\mathbf{x}_0)| \geq a_k > 0. \tag{3.19}$$

   where a_k depends only on k and n. This is because the set $\{(\xi \cdot \nabla)^k\}_{|\xi|=1}$ forms an overcomplete basis for k-th order derivatives.

2. *Step 2: Localization and Coordinate Alignment:*

   2.1 **Neighborhood Selection**: As we know, the k-th derivative $(\xi \cdot \nabla)^k \phi(\mathbf{x})$ is continuous in $\mathbf{x}$ (since ϕ is smooth enough). By continuity, for any $\varepsilon > 0$, there exists a neighborhood

$B(\mathbf{x_0}, r)$ where:

$$|(\xi \cdot \nabla)^k \phi(\mathbf{x}) - (\xi \cdot \nabla)^k \phi(\mathbf{x_0})| < \varepsilon. \tag{3.20}$$

We choose $\varepsilon = a_k/2$, then:

$$|(\xi \cdot \nabla)^k \phi(\mathbf{x})| \geq |(\xi \cdot \nabla)^k \phi(\mathbf{x_0})| - \varepsilon \geq a_k - a_k/2 = a_k/2. \tag{3.21}$$

Thus, we have:

$$|(\xi \cdot \nabla)^k \phi(\mathbf{x})| \geq a_k/2 \quad \forall \mathbf{x} \in B(\mathbf{x}_0, r). \tag{3.22}$$

2.2 **Finite Cover Construction**: Since $\operatorname{supp}(\psi)$ is compact, select a finite subcover $\{B_j\}_{j=1}^M$ with associated directions $\{\xi^j\}_{j=1}^M$ and partition of unity $\{\eta_j\}_{j=1}^M$ subordinate to $\{B_j\}$.

3. *Step 3: Iterated Integration and 1D Reduction:*

3.1 **Coordinate Rotation**: In each B_j, perform a rotation R_j aligning ξ^j with $\mathbf{e}_1$. Under the change of variables $\mathbf{y} = R_j\mathbf{x}$:

$$(\xi^j \cdot \nabla_{\mathbf{x}})^k = \partial_{y_1}^k. \tag{3.23}$$

3.2 **1D van der Corput Application**: The integral over B_j becomes:

$$\int_{B_j} e^{i\lambda\phi(\mathbf{x})} \psi(\mathbf{x})\, d\mathbf{x} = \int e^{i\lambda\phi(R_j^{-1}\mathbf{y})} \tilde{\psi}_j(\mathbf{y})\, d\mathbf{y}. \tag{3.24}$$

where $\tilde{\psi}_j(\mathbf{y}) = \psi(R_j^{-1}\mathbf{y})\eta_j(R_j^{-1}\mathbf{y})|\det R_j^{-1}|$. Applying Fubini's theorem:

$$\int_{\mathbb{R}^{n-1}} \left(\int_{\mathbb{R}} e^{i\lambda\phi(R_j^{-1}(y_1,\mathbf{y}'))} \tilde{\psi}_j(y_1, \mathbf{y}')\, dy_1 \right) d\mathbf{y}'. \tag{3.25}$$

where we split $\mathbf{y}$ into y_1 (first coordinate) and $\mathbf{y}'$ (remianing $n-1$ coordinates). For each fixed $\mathbf{y}'$, compute the 1D integral over y_1. Then, we integrate over $\mathbf{y}'$ by combining the results by integrating over $\mathbb{R}^{n-1}$.

3.3 **Uniform Estimate**: For fixed $\mathbf{y}'$, the inner integral has phase $\phi_{y'}(y_1) := \phi(R_j^{-1}(y_1, \mathbf{y}'))$ satisfying $|\partial_{y_1}^k \phi_{y'}(y_1)| \geq a_k/2$. By the **Prop 2 (1D van der Corput), Ref:** (2.2) [12]:

$$\left| \int_{\mathbb{R}} e^{i\lambda\phi_{y'}(y_1)} \tilde{\psi}_j(y_1, \mathbf{y}')\, dy_1 \right| \leq C_k (a_k\lambda)^{-1/k} \left(\|\tilde{\psi}_j(\cdot, \mathbf{y}')\|_{L^\infty} + \|\partial_{y_1}\tilde{\psi}_j(\cdot, \mathbf{y}')\|_{L^1} \right). \tag{3.26}$$

Conclusion: Global Estimate Integrating (3.26) over $\mathbf{y}'$ and summing over j yields:

$$|I(\lambda)| \leq \sum_{j=1}^M C_k (a_k\lambda)^{-1/k} \left(\|\psi\|_{L^\infty} + \|\nabla\psi\|_{L^1} \right) \leq C_k' \lambda^{-1/k} \left(\|\psi\|_{L^\infty} + \|\nabla\psi\|_{L^1} \right). \tag{3.27}$$

[12] This version of the lemma states that if $|\phi^{(k)}(x)| \geq 1$ and ψ is integrable with an integrable derivative, then $\left| \int e^{i\lambda\phi(x)} \psi(x) dx \right| \leq C_k \lambda^{-1/k} \left(|\psi(b)| + \int |\psi'(x)| dx \right)$. The norms $\|\tilde{\psi}_j(\cdot, \mathbf{y}')\|_{L^\infty}$ and $\|\partial_{y_1}\tilde{\psi}_j(\cdot, \mathbf{y}')\|_{L^1}$ are the natural generalizations of these terms for an amplitude defined on $\mathbb{R}$. The constant C_k absorbs the factor from the lower bound $a_k/2$.

where C'_k depends on k, n, the number of patches M, and the minimal a_k.

□

3.2.2 *Directional Powers Span*

We now prove the assertion used in the main proof of Proposition 5: that the homogeneous polynomials of the form $(\xi \cdot x)^k$ span the entire space of homogeneous polynomials of degree k.

1. *Step 1: Setup and Goal:*

   1.1 **The Space:** Let V be the real linear space of homogeneous polynomials of degree k in $\mathbb{R}^n$. Its dimension is $d(k,n) = \binom{n+k-1}{k}$. A standard basis is the set of monomials $\{x^\alpha : |\alpha| = k\}$.

   1.2 **The Claim:** We claim there exists a finite set of unit vectors $\{\xi^1, \dots, \xi^{d(k,n)}\} \subset S^{n-1}$ such that the polynomials

   $$P_j(x) = (\xi^j \cdot x)^k, \qquad j = 1, \dots, d(k,n). \tag{3.28}$$

   also form a basis for V.

2. *Step 2: Define an Inner Product and a Key Property:*

   2.1 **Inner Product:** For any two polynomials $P(x) = \sum_{|\alpha|=k} a_\alpha x^\alpha$ and $Q(x) = \sum_{|\alpha|=k} b_\alpha x^\alpha$ in V, define the inner product:

   $$\langle P, Q \rangle = \sum_{|\alpha|=k} \alpha!\, a_\alpha b_\alpha. \tag{3.29}$$

   This is a positive definite inner product on V.

   2.2 **Crucial Property:** This inner product has the following important property:

   $$\langle P, Q \rangle = \left[Q\left(\frac{\partial}{\partial x}\right) \right](P). \tag{3.30}$$

   where the right-hand side means applying the differential operator $Q(\partial/\partial x)$ to the polynomial $P(x)$ and evaluating the result (which is a constant).

3. *Step 3: Proof by Contradiction:*

   3.1 **Assume the Negation:** Assume for contradiction that the set $\{(\xi \cdot x)^k : \xi \in \mathbb{R}^n\}$ does *not* span V. Then there must exist a non-zero polynomial $P \in V$ that is orthogonal to every polynomial of this form. That is, for all $\xi \in \mathbb{R}^n$:

   $$\langle P, (\xi \cdot \cdot)^k \rangle = 0. \tag{3.31}$$

3.2 **Apply the Key Property:** Using the property from (3.30), this orthogonality condition implies:

$$\left[(\xi \cdot \nabla)^k\right](P) = 0 \quad \text{for all } \xi \in \mathbb{R}^n. \tag{3.32}$$

3.3 **Interpret the Condition:** (3.32) means that for every fixed direction ξ, the k-th directional derivative of P is identically zero. In other words, for every $\xi \in \mathbb{R}^n$, the single-variable function

$$f_\xi(t) = P(t\xi) \tag{3.33}$$

is a polynomial in t of degree less than k[13].

3.4 **Use Homogeneity:** However, since P is homogeneous of degree k[14], we know that:

$$P(t\xi) = t^k P(\xi). \tag{3.34}$$

The only way for $t^k P(\xi)$ to be a polynomial in t of degree less than k for all ξ is if

$$P(\xi) = 0 \quad \text{for all } \xi \in \mathbb{R}^n. \tag{3.35}$$

3.5 **Reach a Contradiction:** Therefore, P must be the zero polynomial. This contradicts our initial assumption that P was non-zero.

4. *Step 4: Conclusion:* Since our assumption led to a contradiction, the opposite must be true: the set $\{(\xi \cdot x)^k : \xi \in \mathbb{R}^n\}$ spans the space V. Consequently, a finite basis of this form must exist. This completes the proof that we can choose unit vectors $\xi^1, \ldots, \xi^{d(k,n)}$ such that the polynomials $(\xi^j \cdot x)^k$ form a basis.

Remarks on Proposition 5:

1. **Sharpness of the Estimate.** The estimate (3.17) is not always optimal. For instance, in $\mathbf{R}^2$ with $\phi(x) = x_1 x_2$, the estimate gives $\mathcal{O}(\lambda^{-1/2})$ decay, while the true decay is actually $\mathcal{O}(\lambda^{-1})$.

2. **Canonical Forms and Dimension.** In one dimension, any phase with a critical point where the derivative doesn't vanish to infinite order can be transformed into the canonical form $\pm x^k$ near that point. This powerful simplification has no direct analogue in higher dimensions, except for the special case of nondegenerate critical points ($k = 2$), which we now consider in the next section.

[13] If the k-th derivative of a function is identically zero on $\mathbb{R}$, and the function is sufficiently smooth (which a polynomial is), then the function must be a polynomial of degree at most $k-1$. Here, $\frac{d^k}{dt^k} f_\xi(t) = (\xi \cdot \nabla)^k P(t\xi) = 0$.

[14] A polynomial P is homogeneous of degree k if every one of its terms has the same total degree k. This means it satisfies the scaling law: $P(tx_1, \ldots, tx_n) = t^k P(x_1, \ldots, x_n)$ for any scalar t.

3.3 Asymptotics: Nondegenerate Critical Points

Intuition: Near nondegenerate critical points, the phase function ϕ can be approximated by its quadratic Taylor expansion. The Hessian matrix's determinant governs the interference pattern, producing the characteristic $\lambda^{-n/2}$ decay. The signature (number of positive/negative eigenvalues) determines the oscillatory behavior through the complex phase factor $e^{i\pi \mathrm{sgn}(H)/4}$.

[Proposition 6: Stationary Phase in $\mathbb{R}^n$] Suppose:

i. $\nabla\phi(\mathbf{x}_0) = \mathbf{0}$ (critical point)

ii. $\det H\phi(\mathbf{x}_0) \neq 0$ (nondegenerate Hessian)

iii. $\psi \in C_c^\infty(\mathbb{R}^n)$ supported near $\mathbf{x}_0$

Then the oscillatory integral admits the asymptotic expansion:

$$I(\lambda) = \int_{\mathbb{R}^n} e^{i\lambda\phi(x)}\psi(x)dx \sim \lambda^{-n/2} e^{i\lambda\phi(\mathbf{x}_0)} \sum_{j=0}^{\infty} a_j \lambda^{-j}, \quad \text{as } \lambda \to \infty. \tag{3.36}$$

where the coefficients a_j depend on ψ and derivatives of ϕ at $\mathbf{x}_0$.

We proceed in three steps, beginning with the explicit computation for quadratic phase functions.

1. *Step 1: Exact Gaussian Integrals for Quadratic Forms:* We have the below observation.

   [1st Observation: Quadratic Phase Reduction] Let $Q(x)$ be the normalized quadratic form:

   $$Q(x) = \sum_{j=1}^{m} x_j^2 - \sum_{j=m+1}^{n} x_j^2, \quad 0 \leq m \leq n. \tag{3.37}$$

   where m counts the positive eigenvalues of the Hessian. Following the method in Proposition 3, we evaluate:

   $$\int_{\mathbb{R}^n} e^{i\lambda Q(x)} e^{-|x|^2} x^\ell \, dx \sim \lambda^{-n/2-|\ell|/2} \sum_{j=0}^{\infty} c_j(m,\ell)\lambda^{-j}. \tag{3.38}$$

   where $\ell = (\ell_1, \ldots, \ell_n)$ is a multi-index and $x^\ell = x_1^{\ell_1} \cdots x_n^{\ell_n}$. Moreover, if any ℓ_j is odd, the integral vanishes by symmetry.

   Proof. To prove the above, we can do the following:

   1.1 **Factorization into 1D Integrals:**

1.1.1 **The Quadratic Form:** The 3.37 can be written as:

$$Q(x) = \sum_{j=1}^{m}(+1)x_j^2 + \sum_{j=m+1}^{n}(-1)x_j^2 = \sum_{j=1}^{n}\sigma_j x_j^2 = \frac{1}{2}x^T Hx. \tag{3.39}$$

where H is the Hessian and σ_j explicitly track whether each direction contributes a "+" or "−" to the phase. The Hessian matrix H is computed via second derivatives:

$$H_{ij} = \frac{\partial^2 Q}{\partial x_i \partial x_j} = \begin{cases} 2 & \text{if } i = j \text{ and } 1 \le j \le m, \\ -2 & \text{if } i = j \text{ and } m+1 \le j \le n, \\ 0 & \text{if } i \ne j. \end{cases} \implies H = 2\begin{pmatrix} I_m & 0 \\ 0 & -I_{n-m} \end{pmatrix}. \tag{3.40}$$

where I_k is the $k \times k$ identity matrix, and the σ_j encodes the sign of the eigenvalue for each coordinate:

$$\sigma_j = \begin{cases} +1 & \text{for } 1 \le j \le m \quad \text{(positive eigenvalues)}, \\ -1 & \text{for } m+1 \le j \le n \quad \text{(negative eigenvalues)}. \end{cases} \tag{3.41}$$

So in 3.39, the $\frac{1}{2}$ cancels the 2 from the Hessian, giving back the original $Q(x)$.

1.1.2 **The Integral Factors Expression:** Moreover, since $Q(x)$ is diagonal, the integral factors can be expressed as:

$$\begin{aligned} \int_{\mathbb{R}^n} e^{i\lambda Q(x)} e^{-|x|^2} x^\ell dx &= \int_{\mathbb{R}^n} \underbrace{e^{i\lambda \sum_{j=1}^n \sigma_j x_j^2}}_{①} \underbrace{e^{-\sum_{j=1}^n x_j^2}}_{②} \underbrace{x^\ell}_{③} d_{x_1} \cdots dx_{x_n}. \\ &= \int_{\mathbb{R}^n} \left(\prod_{j=1}^n e^{i\lambda\sigma_j x_j^2}\right)\left(\prod_{j=1}^n e^{-x_j^2}\right)\left(\prod_{j=1}^n x_j^{\ell_j}\right) d_{x_1} \cdots dx_{x_n}. \end{aligned} \tag{3.42}$$

where ①: the exponential of a sum, ②: Gaussian weight, and ③: the monomial term. Since all terms are products of 1D functions and $e^{-x_j^2}$ ensures absoulte convenience, Fubini's Theorem [15]allows us to separate the integrals as:

$$\int_{\mathbb{R}^n} e^{i\lambda Q(x)} e^{-|x|^2} x^\ell dx = \prod_{j=1}^n \left(\int_{-\infty}^{\infty} e^{i\lambda\sigma_j x_j^2} e^{-x_j^2} x_j^{\ell_j} d_{x_j}\right). \tag{3.43}$$

1.2 **Evaluation of 1D Integrals:** For each j, we consider:

$$I_j = \int_{-\infty}^{\infty} e^{-(1-i\lambda\sigma_j)x_j^2} x_j^{\ell_j} dx_j. \tag{3.44}$$

Here, we have two cases:

[15]Fubini's Theorem justifies: $\int_{\mathbb{R}^n}\left(\prod_{j=1}^n f_j(x_j)\right) d_{x_1}\cdots d_{x_n} = \prod_{j=1}^n \left(\int_{-\infty}^{\infty} f_j(x_j) d_{x_j}\right)$.

i. **Case i: ℓ_j odd:** The integrand is odd, so $I_j = 0$.

ii. **Case ii: ℓ_j even:**

1.2.1 **Evenness of the Integrand:** Let $k = \ell_j$, we then notice that since x_j^k is even and $e^{-(1-i\lambda\sigma_j)x_j^2}$ is also even, we can do the below simplification:

$$I_j = \int_{-\infty}^{\infty} e^{-(1-i\lambda\sigma_j)x_j^2} x_j^k d_{x_j} \implies I_j = 2\int_0^{\infty} e^{-(1-i\lambda\sigma_j)x_j^2} x_j^k d_{x_j}. \tag{3.45}$$

1.2.2 **Change of Variable:** We let $u = x_j^2$, and $du = 2x_j d_{x_j}$, then the integral becomes:

$$I_j = 2\int_0^{\infty} e^{-(1-i\lambda\sigma_j)u} u^{k/2} \frac{du}{2\sqrt{u}} = \int_0^{\infty} e^{-(1-i\lambda\sigma_j)u} u^{(k-1)/2} du. \tag{3.46}$$

1.2.3 **The Gamma Integral:** The integral now matches the Gamma function [16] form. We let $t = (1 - i\lambda\sigma_j)u$ and $dt = (1 - i\lambda\sigma_j)du$, we then have:

$$I_j = \int_0^{\infty} e^{-t} \left(\frac{t}{1-i\lambda\sigma_j}\right)^{(k-1)/2} \frac{dt}{1-i\lambda\sigma_j} = (1-i\lambda\sigma_j)^{-(k+1)/2} \int_0^{\infty} e^{-t} t^{(k-1)/2} dt. \tag{3.47}$$

The remaining integral is $\Gamma\left(\frac{k+1}{2}\right)$:

$$I_j = \Gamma\left(\frac{k+1}{2}\right)(1-i\lambda\sigma_j)^{-(k+1)/2}. \tag{3.48}$$

1.3 **Asymptotic Expansion for Large λ:**

1.3.1 **Expand $(1 - i\lambda\sigma_j)^{-(k+1)/2}$:**

$$(1-i\lambda\sigma_j)^{-(k+1)/2} = \underbrace{(-\sigma_j i\lambda)^{-(k+1)/2}}_{①} \underbrace{\left(1 + \frac{i}{\sigma_j\lambda}\right)^{-(k+1)/2}}_{②} \tag{3.49}$$

1.3.2 **The dominant term comes from ①:**

$$(-\sigma_j i\lambda)^{-(k+1)/2} = \lambda^{-(k+1)/2}\left(e^{i\pi(1-\sigma_j)/2}\right)^{-(k+1)/2} = \lambda^{-(k+1)/2} e^{i\pi\sigma_j(k+1)/4}. \tag{3.50}$$

where $i = e^{i\pi/2}$ and $-\sigma_j = e^{i\pi(1-\sigma_j)/2}$(because $\sigma_j = \pm 1$).

1.3.3 **Binomial Expansion for ②:** For large λ, $|\frac{i}{\sigma_j\lambda}| \ll 1$, we can use the generalized binomial series [17], With $x = \frac{i}{\sigma_j\lambda}$ and $\alpha = -\frac{k+1}{2}$, we have:

$$\left(1 + \frac{i}{\sigma_j\lambda}\right)^{-(k+1)/2} = 1 - \frac{(k+1)}{2}\cdot\frac{i}{\sigma_j\lambda} + \frac{(k+1)(k+3)}{8}\cdot\frac{1}{\sigma_j^2\lambda^2} + \cdots. \tag{3.51}$$

[16] The Gamma function form is: $\Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt, \quad \mathrm{Re}(z) > 0.$

1.3.4 **Combine Results:** We combine the results from ① and ②.

$$(1 - i\lambda\sigma_j)^{-(k+1)/2} = (-\sigma_j i\lambda)^{-(k+1)/2} \left(1 + \frac{k+1}{2i\lambda\sigma_j} + \cdots\right)$$
$$= \lambda^{-(k+1)/2} e^{i\pi\sigma_j(k+1)/4} \left(1 - \frac{i(k+1)}{2\lambda\sigma_j} + \cdots\right). \tag{3.52}$$

Substitute 3.52 back with the Gamma expression, we then have:

$$I_j \sim \lambda^{-(k+1)/2} e^{i\pi\sigma_j(k+1)/4} \Gamma\left(\frac{k+1}{2}\right) \sum_{p=0}^{\infty} d_p(k, \sigma_j) \lambda^{-p}. \tag{3.53}$$

where d_p are coefficients from the binomial expansion.

1.4 **Combine All Dimensions:**

From step 1.3, each 1D Integral I_j has the asymptotic expansion as 3.53. Moreover, with $(k+1)/2 = \frac{\ell_j+1}{2}$, it can also be expressed as:

$$I_j \sim \lambda^{-(\ell_j+1)/2} e^{i\pi\sigma_j(\ell_j+1)/4} \Gamma\left(\frac{\ell_j+1}{2}\right) \sum_{p=0}^{\infty} d_p(\ell_j, \sigma_j) \lambda^{-p}. \tag{3.54}$$

The n-dimensional Integral is the product of these 1D Integrals:

$$\prod_{j=1}^{n} I_j \sim \prod_{j=1}^{n} \left(\lambda^{-(\ell_j+1)/2} e^{i\pi\sigma_j(\ell_j+1)/4} \Gamma\left(\frac{\ell_j+1}{2}\right) \sum_{p=0}^{\infty} d_p(\ell_j, \sigma_j) \lambda^{-p}\right). \tag{3.55}$$

Now, we factor out the $\lambda-$ dependence:

$$\prod_{j=1}^{n} I_j \sim \underbrace{\left(\prod_{j=1}^{n} \lambda^{-(\ell_j+1)/2}\right)}_{①} \underbrace{\left(\prod_{j=1}^{n} e^{i\pi\sigma_j(\ell_j+1)/4} \Gamma\left(\frac{\ell_j+1}{2}\right)\right)}_{②} \underbrace{\left(\prod_{j=1}^{n} \sum_{p=0}^{\infty} d_p(\ell_j, \sigma_j) \lambda^{-p}\right)}_{③}. \tag{3.56}$$

1.4.1 **Simplify the λ-Exponent in ① :** We have $|\ell| = \sum_{j=1}^{n} \ell_j$, which gives us:

$$\sum_{j=1}^{n} \frac{\ell_j + 1}{2} = \frac{1}{2}\left(\sum_{j=1}^{n} \ell_j + \sum_{j=1}^{n} 1\right) = \frac{|\ell| + n}{2}. \tag{3.57}$$

Thus, we can have the below expression:

$$\prod_{j=1}^{n} \lambda^{-(\ell_j+1)/2} = \lambda^{-\sum_{j=1}^{n}(\ell_j+1)/2} = \lambda^{-(n+|\ell|)/2}. \tag{3.58}$$

[17] The generalized binomial series: $(1+x)^{\alpha} = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots$ (for $|x| < 1$).

1.4.2 **Analyze the Phase and Gamma terms in ②:**

$$\prod_{j=1}^{n} e^{i\pi\sigma_j(\ell_j+1)/4} = e^{i\pi\sum_{j=1}^{n}\sigma_j(\ell_j+1)/4}, \quad \prod_{j=1}^{n}\Gamma\left(\frac{\ell_j+1}{2}\right) = \text{Constants}. \tag{3.59}$$

where the constants depending on ℓ.

1.4.3 **Asymptotic Series Multiplication in ③:** The product of n asymptotic series is:

$$\prod_{j=1}^{n} S_j(\lambda), \quad \text{where } S_j(\lambda) = \sum_{p=0}^{\infty} d_p(\ell_j,\sigma_j)\lambda^{-p}. \tag{3.60}$$

is expanded via multinomial multiplication, yielding coefficients determined by:

i. General Coefficient Formula: For each order $q \geq 0$:

$$\tilde{c}_q(m,\ell) = \sum_{\substack{(p_1,\ldots,p_n)\in\mathbb{N}_0^n \\ |\mathbf{p}|=q}} \prod_{j=1}^{n} d_{p_j}(\ell_j,\sigma_j). \tag{3.61}$$

where $|\mathbf{p}| = p_1 + \cdots + p_n$.

ii. Leading Order ($q = 0$):

$$\tilde{c}_0 = \prod_{j=1}^{n} d_0(\ell_j,\sigma_j) = 1. \tag{3.62}$$

since $d_0(\ell_j,\sigma_j) = 1$ for all j (from the binomial expansion).

iii. First Order ($q = 1$):

$$\tilde{c}_1 = \sum_{j=1}^{n} d_1(\ell_j,\sigma_j) = -\frac{i}{2}\sum_{j=1}^{n}\frac{\ell_j+1}{\sigma_j}. \tag{3.63}$$

where each $d_1(\ell_j,\sigma_j) = -\frac{i(\ell_j+1)}{2\sigma_j}$.

iv. Second Order ($q = 2$): Contains two contributions:

$$\tilde{c}_2 = \underbrace{\sum_{j=1}^{n} d_2(\ell_j,\sigma_j)}_{\text{single } p_j=2 \text{ terms}} + \underbrace{\sum_{1\leq j<k\leq n} d_1(\ell_j,\sigma_j)d_1(\ell_k,\sigma_k)}_{\text{pairwise } p_j=p_k=1 \text{ terms}}. \tag{3.64}$$

Note: The coefficients $\tilde{c}_q$ incorporate interdimensional coupling, Hessian signature (σ_j), and ℓ-dependence.

1.4.4 Combine All Terms: Assembling the components from Steps 1.4.1–1.4.3:

$$\prod_{j=1}^{n} I_j \sim \lambda^{-\frac{n+|\ell|}{2}} \underbrace{\prod_{j=1}^{n} e^{i\pi\sigma_j(\ell_j+1)/4}}_{\text{Phase}} \underbrace{\prod_{j=1}^{n}\Gamma\left(\frac{\ell_j+1}{2}\right)}_{\text{Gamma factors}} \underbrace{\sum_{q=0}^{\infty}\tilde{c}_q(m,\ell)\lambda^{-q}}_{\text{Asymptotic series}}. \tag{3.65}$$

1.5 **Final Asymptotic Form:** We know that the signature of Q (number of positive minus negative eigenvalues) emerges from the phase factors, which can be seen as:

$$\sum_{j=1}^{n} \sigma_j(\ell_j + 1) = \operatorname{sgn}(Q) + \sum_{j=1}^{n} \sigma_j \ell_j, \quad \text{where } \operatorname{sgn}(Q) = 2m - n. \tag{3.66}$$

For the dominant term ($\ell = 0$), we have the phase simplifies to $e^{i\pi \operatorname{sgn}(Q)/4}$. Moreover, we can absorb all constants into $c_j(m, \ell)$ to obtain the compact form:

$$\int_{\mathbb{R}^n} e^{i\lambda Q(x)} e^{-|x|^2} x^\ell dx \sim \lambda^{-\frac{n+|\ell|}{2}} e^{i\pi \operatorname{sgn}(Q)/4} \sum_{j=0}^{\infty} c_j(m, \ell) \lambda^{-j}. \tag{3.67}$$

□

2. *Step 2: Localization via Conical Partition:*

[2nd Observation: Key Decay Estimate] For any $\eta \in C_0^\infty(\mathbb{R}^n)$ and multi-index ℓ, the oscillatory integral satisfies:

$$\left| \int_{\mathbb{R}^n} e^{i\lambda Q(x)} x^\ell \eta(x)\, dx \right| \leq C_{n,\ell} \lambda^{-n/2 - |\ell|/2}. \tag{3.68}$$

where $C_{n,\ell}$ depends only on dimension n and ℓ.

Proof. We proceed in three steps through decomposing the integral using a conical partition of unity and integrate by parts.

2.2 **Cone Decomposition:**

2.2.1 **Cone Geometry:** For each coordinate direction $j = 1, \ldots, n$, we can define:

$$\Gamma_j := \left\{ x \in \mathbb{R}^n : |x_j| \geq \frac{|x|}{\sqrt{2n}} \right\}, \quad \Gamma_j^0 := \left\{ x : |x_j| \geq \frac{|x|}{\sqrt{n}} \right\}. \tag{3.69}$$

These conditions compare the j-th coordinate $|x_j|$ to the Euclidean norm $|x| = \sqrt{x_1^2 + \cdots + x_n^2}$. Moreover, these satisfy that $\Gamma_j^0 \subset \Gamma_j$ (strictly smaller aperture), $\bigcup_{j=1}^n \Gamma_j^0 = \mathbb{R}^n \setminus \{0\}$ (covering property), and in Γ_j, the j-th coordinate dominates: $|x_j| \geq (2n)^{-1/2}|x|$.

2.2.2 **Partition of Unity:** Construct functions $\{\Omega_j\}_{j=1}^n$ such that:

$$\sum_{j=1}^{n} \Omega_j(x) = 1 \ \ \forall x \neq 0, \quad \operatorname{supp}(\Omega_j) \subset \Gamma_j, \quad \Omega_j \in C^\infty(\mathbb{R}^n \setminus \{0\}). \tag{3.70}$$

with Ω_j homogeneous of degree 0 such that $\Omega_j(tx) = \Omega_j(x)$ for $t > 0$.

2.2.3 **Localization:** For $\eta \in C_0^\infty(\mathbb{R}^n)$, we decompose the integral:

$$\begin{aligned}\int_{\mathbb{R}^n} e^{i\lambda Q(x)} x^\ell \eta(x) dx &= \sum_{j=1}^n \int_{\mathbb{R}^n} e^{i\lambda Q(x)} x^\ell \eta(x)\Omega_j(x) dx \\ &= \sum_{j=1}^n \int_{\Gamma_j} e^{i\lambda Q(x)} x^\ell \underbrace{\eta(x)\Omega_j(x)}_{\text{Supported in } \Gamma_j} dx. \end{aligned} \tag{3.71}$$

where $\Omega_j(x) = 0$ outside Γ_j, and the inetgral over $\mathbb{R}^n$ reduces to an integral over Γ_j.

2.2 **Integration by Parts:**

2.2.1 **The Differential Operator:** On each Γ_j, we use the differential operator:

$$D_j = \frac{1}{2i\lambda\sigma_j x_j}\frac{\partial}{\partial x_j}, \quad \text{where } Q(x) = \sum_{k=1}^n \sigma_k x_k^2 \ \ (\sigma_k \pm 1). \tag{3.72}$$

with the key property:

$$D_j e^{i\lambda Q(x)} = \frac{1}{2i\lambda\sigma_j x_j}\cdot\frac{\partial}{\partial x_j}e^{i\lambda Q(x)} = \frac{1}{2i\lambda\sigma_j x_j}\cdot e^{i\lambda Q(x)}\cdot 2i\lambda\sigma_j x_j = e^{i\lambda Q(x)}. \tag{3.73}$$

where the operator D_j is designed to "peel off" the exponential term $e^{i\lambda Q(x)}$ when applied, leaving the rest of the integrand untouched.

2.2.2 **Integration by parts:** Now, we do integration by parts. For a locallized inetgral on Γ_j:

$$\int_{\Gamma_j} e^{i\lambda Q(x)} F(x) dx = \int_{\Gamma_j} (D_j e^{i\lambda Q(x)}) F(x) dx. \tag{3.74}$$

where $F(x) = x^\ell \eta(x)\Omega_j(x)$. Now integrate by parts in x_j, we have:

$$I = \underbrace{\left[e^{i\lambda Q(x)}\cdot\left(\frac{F(x)}{2i\lambda\sigma_j x_j}\right)\right]_{\partial\Gamma_j}}_{:=0 \text{ due to } \eta\in C_0^\infty(\mathbb{R}^n)} - \int_{\Gamma_j} e^{i\lambda Q(x)}\partial_{x_j}\left(\frac{F(x)}{2i\lambda\sigma_j x_j}\right) dx. \tag{3.75}$$

The boundary terms vanish. We can also represent 3.75 in terms of its adjoint, where its adjoint ${}^tD_j$ is:

$${}^tD_j F(x) = -\frac{\partial}{\partial x_j}\left(\frac{F(x)}{2i\lambda\sigma_j x_j}\right). \tag{3.76}$$

After IBP of N times, we have:

$$({}^tD_j)^N F(x) = \left(-\frac{\partial}{\sigma_j x_j \partial x_j}\right)^N \frac{F(x)}{(2i\lambda)^N} = \frac{(-1)^N}{(2i\lambda)^N}\left(\frac{\partial}{\sigma_j x_j\partial x_j}\right)^N F(x). \tag{3.77}$$

We can then substitute 3.77 into 3.75, this then gives us:

$$\left|\int_{\Gamma_j} e^{i\lambda Q(x)}F(x)dx\right| = \left|\int_{\Gamma_j} e^{i\lambda Q(x)}({}^tD_j)^N F(x)dx\right|.$$
$$\leq \int_{\Gamma_j} \underbrace{\left|e^{i\lambda Q(x)}\right|}_{|e^{i\lambda Q(x)}|=1} \cdot \left|({}^tD_j)^N F(x)\right| dx = \int_{\Gamma_j} \left|\frac{(-1)^N}{(2i\lambda)^N}\left(\frac{\partial}{\sigma_j x_j \partial x_j}\right)^N F(x)\right| dx. \tag{3.78}$$

Because $|i^N| = 1$ and $|(-1)^N| = 1$. Thus:

$$\left|\int_{\Gamma_j}\right| \leq (2\lambda)^{-N}\int_{\Gamma_j}\left|\left(\frac{\partial}{\sigma_j x_j \partial x_j}\right)^N \underbrace{[x^\ell \eta(x)\Omega_j(x)]}_{F(x)}\right| dx. \tag{3.79}$$

2.3 **Decay Estimation:** Now, we want to analyze the integrand from 3.79 for a decay estimate.

2.3.1 **Controlling the Derivatives:** We have the integrand as:

$$\left(\frac{\partial}{\sigma_j x_j \partial x_j}\right)^N [x^\ell \eta(x)\Omega_j(x)]. \tag{3.80}$$

and we have the below observations:

i. Dominant Term: For $F(x) = x^\ell \eta(x)\Omega_j(x)$, the worst-case term after applying $\left(\frac{\partial}{\sigma_j x_j \partial x_j}\right)^N$ comes from:

$$\left(\frac{\partial}{\partial x_j}\right)^N x^\ell = \begin{cases} \frac{\ell_j!}{(\ell_j - N)!}x^{\ell - Ne_j} & \text{if } \ell_j \geq N, \\ 0 & \text{if } \ell_j \leq N. \end{cases} \tag{3.81}$$

where $e_j = (0, \cdots, 1, \cdots, 0)$ (1 in the j-th position). This term is dominant, because differentiating $x_j^{\ell_j}$ repeatedly gives the largest power of x left over.

ii. Homogeneity of Ω_j: Since Ω_j is homogeneous of degree 0, its derivatives scale like:

$$|\partial_{xk}\Omega_j(x)| \leq C|x|^{-1}. \tag{3.82}$$

Thus, derivatives of Ω_j introduce factors of $|x|^{-1}$.

iii. Compact Support of η: $\eta \in C_0^\infty$ ensures all derivatives of η are bounded.

2.3.2 **Power Counting:** On Γ_j, $|x_j| \sim |x|$, so $|x^{\ell - Ne_j}| \sim |x|^{|\ell| - N}$. We now do integration in polar coordinates:

$$\int_{\Gamma_j} |x|^{|\ell|-N} dx \sim \int_0^R r^{|\ell|-N+n-1} dr. \tag{3.83}$$

For convergence, we need $|\ell| - N + n < 0$ (e.g., $N > n + |\ell|$). The integral bound becomes:

$$\left|\int_{\Gamma_j}\right| \leq C\lambda^{-N} \cdot R^{|\ell|-N+n} \implies \left|\int_{\Gamma_j}\right| \leq C'\lambda^{-n/2-|\ell|/2}. \tag{3.84}$$

To optimize, we choose $N = n + |\ell| + 1$ such that $\lambda^{-N}R^{|\ell|-N+n} \sim \lambda^{-n/2-|\ell|/2}$.

□

3. *Step 3: Morse lemma reduction:* To extend the asymptotic formula to general nondegenerate phases $\phi(x)$, we use Morse's Lemma to transform ϕ into a quadratic form near its critical point as the following:

> **[3rd Observation: Morse's Lemma]** Let $\phi \in C^\infty(R^n)$ with $\phi(0) = \nabla\phi(0) = 0$, and suppose the Hessian $\left[\frac{\partial^2\phi}{\partial x_i \partial x_j}(0)\right]$ is nondegenerate. Then there exists a local diffeomorphism $y = \Psi(x)$ near 0 such that:
>
> $$\phi(x) = \sum_{j=1}^{m} y_j^2 - \sum_{j=m+1}^{n} y_j^2. \tag{3.85}$$
>
> where m is the index of the Hessian.

Proof. We proceed by induction:

3.1 **Taylor Expansion**:

3.1.1 **General Multivariate Taylor Expansion:** For $\phi \in C^\infty(\mathbb{R}^n)$, the Taylor series expansion around $a \in \mathbb{R}^n$ is:

$$\phi(x) = \sum_{|\alpha|\geq 0} \frac{D^\alpha\phi(a)}{\alpha!}(x-a)^\alpha. \tag{3.86}$$

where $\alpha = (\alpha_1, \cdots, \alpha_n)$ is a multi-index ($|\alpha| = \alpha_1 + \cdots + \alpha_n$), $D^\alpha\phi = \frac{\partial^{|\alpha|}\phi}{\partial_{x_1}^{\alpha_1}\cdots\partial_{x_n}^{\alpha_n}}$, $(x-a)^\alpha = (x_1-a_1)^{\alpha_1}\cdots(x_n-a_n)^{a_n}$, and $\alpha! = \alpha_1!\cdots\alpha_n!$. Expand this:

$$\begin{aligned}\phi(x) = \phi(\alpha) + \sum_{i=1}^{n}\frac{\partial\phi}{\partial x_i}(a)(x_i-a_i) + \frac{1}{2}\sum_{i,j=1}^{n}\frac{\partial^2\phi}{\partial x_i\partial x_j}(a)(x_i-a_i)(x_j-a_j) \\ + \sum_{|\alpha|\geq 3}\frac{D^\alpha\phi(a)}{\alpha!}(x-a)^\alpha.\end{aligned} \tag{3.87}$$

3.1.2 **Special Case:** Now, we set $\alpha = 0$, the expansion simplifies to:

$$\phi(x) = \phi(0) + \underbrace{\sum_{i=1}^{n}\frac{\partial\phi}{\partial x_i}(0)x_i}_{:=0} + \frac{1}{2}\sum_{i,j=1}^{n}\frac{\partial^2\phi}{\partial x_i\partial x_j}(0)x_ix_j + \cdots. \tag{3.88}$$

Where $x = 0$ is a critical point ($\nabla\phi(0) = 0$), and the linear term vanishes. Now, we rewrite this using smooth coefficiens $\phi_{ij}(x)$:

$$\phi(x) = \sum_{i,j=1}^{n} x_i x_j \phi_{ij}(x), \quad \phi_{ij}(0) = \frac{1}{2}\frac{\partial^2\phi(0)}{\partial x_i \partial x_j} = \frac{1}{2}\partial^2_{x_i x_j}\phi(0). \tag{3.89}$$

which can be derived via integration (next step). This is equivalent to the standard Taylor expansion but preserves smoothness for all x.

3.2 **Derivation of the Refined Form via Integration:** We have a smooth function $\phi : \mathbb{R}^n \to \mathbb{R}$ with $\phi(0) = 0$ and $\nabla\phi(0) = 0$ ($x = 0$ is a critical point). We want to express $\phi(x)$ purely in terms of second derivatives (Hessian) to isolate the quadratic behavior near $x = 0$.

3.2.1 **First Integration:** From the Fundamental Theorem of Calculus (FTC), we have:

$$\phi(x) - \underbrace{\phi(0)}_{:=0} = \int_0^1 \frac{d}{dt}\phi(tx)dt. \tag{3.90}$$

Here $\phi : \mathbb{R}^n \to \mathbb{R}$ is a smooth scalar function, $x = (x_1, x_2, \cdots, x_n)$ is a fixed vector in $\mathbb{R}^n$, and $t \in \mathbb{R}$ is a scalar parameter. To differentiate $\phi(tx)$, we use multivariable chain rule:

$$\frac{d}{dt}\phi(tx) = \sum_{i=1}^{n} \frac{\partial\phi}{\partial x_i}(tx) \cdot \frac{d(tx_i)}{dt} = \sum_{i=1}^{n} \frac{\partial\phi}{\partial x_i}(tx) \cdot x_i = \nabla\phi(tx) \cdot x. \tag{3.91}$$

Substitute the chain rule result into the integral:

$$\phi(x) = \int_0^1 \frac{d}{dt}\phi(tx)dt = \int_0^1 \nabla\phi(tx) \cdot x dt. \tag{3.92}$$

3.2.2 **Second Integration:** Since $\nabla\phi(0) = 0$, we apply the same FTC to each component of $\nabla\phi$. For each partial derivative $\frac{\partial\phi}{\partial x_i}(tx)$, we define:

$$g_i(s) = \frac{\partial\phi}{\partial x_i}(sx), \quad s \in [0, t]. \tag{3.93}$$

Since $\frac{\partial\phi}{\partial x_i}(0) = 0$ (from $\nabla\phi(0) = 0$), we have:

$$g_i(t) - g_i(0) = \int_0^t g_i'(s)ds \implies \frac{\partial\phi}{\partial x_i}(tx) = \int_0^t \frac{d}{ds}\left(\frac{\partial\phi}{\partial x_i}(sx)\right) ds. \tag{3.94}$$

Using the chain rule to compute the derivative $g_i'(s)$:

$$\frac{d}{ds}\left(\frac{\partial\phi}{\partial x_i}(sx)\right) = \sum_{j=1}^{n} \frac{\partial^2\phi}{\partial x_i \partial x_j}(sx) \cdot \frac{d(sx_j)}{ds} = \sum_{j=1}^{n} \frac{\partial^2\phi}{\partial x_i \partial x_j}(sx) \cdot x_j. \tag{3.95}$$

which is the i-th component of the vector $\text{Hess}(\phi)(sx) \cdot x$, where:

$$\text{Hess}(\phi)(sx) = \left[\frac{\partial^2 \phi}{\partial x_i \partial x_j}(sx) \right]_{i,j=1}^{n}. \tag{3.96}$$

Thus, for each i, we can have:

$$\frac{\partial \phi}{\partial x_i}(tx) = \int_0^t (\text{Hess}(\phi)(sx) \cdot x)_i ds \implies \nabla \phi(tx) = \int_0^t \text{Hess}(sx) \cdot x ds. \tag{3.97}$$

3.2.3 **Substitution Back into $\phi(x)$:** Now, we substitute it back into $\phi(x)$ from (3.92):

$$\phi(x) = \int_0^1 \nabla \phi(tx) \cdot x dt = \int_0^1 \left(\int_0^t \text{Hess}(\phi)(sx) \cdot x ds \right) \cdot x dt. \tag{3.98}$$

Expanding the dot products:

$$\phi(x) = \int_0^1 \int_0^t \sum_{i,j=1}^{n} \frac{\partial^2 \phi}{\partial x_i \partial x_j}(sx) x_i x_j ds dt. \tag{3.99}$$

Now, we swap integrals using Fubini's Theorem. The region $0 \leq s \leq t \leq 1$ is equivalent to $s \in [0,1]$, $t \in [s,1]$. Thus:

$$\int_0^1 \int_0^t f(s) ds dt = \int_0^1 \int_s^1 f(s) dt ds = \int_0^1 (1-s) f(s) ds. \tag{3.100}$$

Applied to $\phi(x)$:

$$\phi(x) = \sum_{i,j=1}^{n} x_i x_j \int_0^1 (1-s) \frac{\partial^2 \phi}{\partial x_i \partial x_j}(sx) ds. \tag{3.101}$$

We now define $\phi_{ij}(x)$:

$$\phi_{ij}(x) = \int_0^1 (1-s) \frac{\partial^2 \phi}{\partial x_i \partial x_j}(sx) ds \implies \phi(x) = \sum_{i,j=1}^{n} x_i x_j \phi_{ij}(x). \tag{3.102}$$

Evaluate at $x = 0$:

$$\phi_{ij}(0) = \int_0^1 (1-s) \frac{\partial^2 \phi}{\partial x_i \partial x_j}(0) ds = \frac{\partial^2 \phi}{\partial x_i \partial x_j}(0) \int_0^1 (1-s) ds = \frac{1}{2} \frac{\partial^2 \phi}{\partial x_i \partial x_j}(0). \tag{3.103}$$

3.3 **Inductive Steps:** From earlier, we define the exact representation.

$$\phi(x) = \sum_{i,j=1}^{n} x_i x_j \phi_{ij}(x), \quad \text{where } \phi_{ij}(0) = \frac{1}{2} \frac{\partial^2 \phi}{\partial x_i x_j}(0). \tag{3.104}$$

which is a quadratic-like form with smooth coefficients $\phi_{ij}(x)$. The goal here is to prove the Morse's Lemma via a smooth decay of coordinates $x \mapsto y$. Assume that after $r-1$

steps, $\phi(y)$ can be written as:

$$\phi(y) = \sum_{j=1}^{r-1} \pm y_j^2 + \sum_{i,j \geq r} y_i y_j \widehat{\phi}_{ij}(y). \tag{3.105}$$

We now eliminate cross-terms involving y_r.

i. **Base Case** $(r = 1)$:

3.3.1 **Taylor Expansion:** We start with the general form 3.104, and we do the Taylor expression.

$$\phi(x) = \sum_{i,j=1}^{n} x_i x_j \phi_{ij}(x) = \underbrace{\phi_{11}(x) x_1^2}_{\text{① pure } x_1^2} + \underbrace{2 \sum_{j=2}^{n} \phi_{1j}(x) x_1 x_j}_{\text{② cross-terms}} + \underbrace{\sum_{i,j=2}^{n} \phi_{ij}(x) x_i x_j}_{\text{③ remaining term}}. \tag{3.106}$$

where $\phi(x) = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i x_j \phi_{ij}(x)$:

i. $(i = 1, j = 1)$: Pure quadratic term $x_1 x_1 \phi_{11}(x) = \phi_{11}(x) x_1^2$.

ii: $(i = 1, j \geq 2)$: Cross-terms with $x_1 \sum_{j=2}^{n} x_1 x_j \phi_{1j}(x)$.

iii: $(i \geq 2, j = 1)$: Symmetric cross-terms $\sum_{i=2}^{n} x_i x_1 \phi_{i1}(x)$.

iv: $(i \geq 2, j \geq 2)$: Remaining quadratic terms $\sum_{i,j=2}^{n} x_i x_j \phi_{ij}(x)$.

Since $\phi_{ij}(x) = \phi_{ji}(x)$ (symmetry of the second derivatives), we then have the combined case ②. Combining these cases yields the expression in (3.106).

3.3.2 **Change of Coordinate:** Now, we do coordinate change. We define the new coordinate as:

$$x_1' = x_1 + \sum_{j=2}^{n} \frac{\phi_{1j}(x)}{\phi_{11}(x)} x_j \implies x_1 = x_1' - \sum_{j=2}^{n} \frac{\phi_{1j}(x)}{\phi_{11}(x)} x_j \quad x_j' = x_j \quad (j \geq 2). \tag{3.107}$$

This transformation isolates x_1^2 by absorbing cross-terms into the new variable x_1'. Moreover, the non-degeneracy condition $\phi_{11}(0) \neq 0$ guarantee the transformation is invertible and smooth near $x = 0$.

3.3.3 **Analysis of the Terms:**

① Pure x_1^2 term:

$$\begin{aligned} \phi_{11}(x) x_1^2 &= \phi_{11}(x) \left(x_1' - \sum_{j=2}^{n} \frac{\phi_{ij}(x)}{\phi_{11}(x)} x_j \right)^2 \\ &= \phi_{11}(x) \left(x_1'^2 - 2 x_1' \sum_{j=2}^{n} \frac{\phi_{ij}(x)}{\phi_{11}(x)} x_j + \left(\sum_{j=2}^{n} \frac{\phi_{ij}(x)}{\phi_{11}(x)} x_j \right)^2 \right). \end{aligned} \tag{3.108}$$

② Cross-terms $x_i x_j$:

$$2\sum_{j=2}^{n}\phi_{ij}x_1x_j = 2\sum_{j=2}^{n}\phi_{ij}(x)\left(x_1' - \sum_{k=2}^{n}\frac{\phi_{1k}(x)}{\phi_{11}(x)}x_k\right)x_j.$$
$$= 2\sum_{j=2}^{n}\phi_{ij}(x)x_1'x_j - 2\sum_{j,k=2}^{n}\frac{\phi_{ij}(x)\phi_{1k}(x)}{\phi_{11}(x)}x_jx_k. \quad (3.109)$$

Now, we combine all the terms together:

$$\phi(x) = \phi_{11}(x)x_1^2 + \phi_{11}(x)\left(\sum_{j=2}^{n}\frac{\phi_{ij}(x)}{\phi_{11}(x)}x_j\right)^2 - 2\sum_{j,k=2}^{n}\frac{\phi_{ij}(x)\phi_{1k}(x)}{\phi_{11}(x)}x_jx_k$$
$$+ \sum_{i,j=2}^{n}\phi_{ij}(x)x_ix_j. \quad (3.110)$$

We now relabel indices $j \to i, k \to j$ and combine terms. The remaining quadratic form becomes:

$$\sum_{i,j=2}^{n}\left(\phi_{ij}(x) - \frac{\phi_{1i}(x)\phi_{1j}(x)}{\phi_{11}(x)}\right)x_ix_j. \quad (3.111)$$

Thus, in the new coordinates, the function takes the desired form:

$$\phi(x') = \underbrace{\phi_{11}(x)x_1'^2}_{\text{isolated } x_1^2} + \sum_{i,j=2}^{n}\underbrace{\left(\phi_{ij}(x) - \frac{\phi_{1i}(x)\phi_{1j}(x)}{\phi_{11}(x)}\right)}_{\text{New coefficients } \widehat{\phi}_{ij}(x)}x_ix_j. \quad (3.112)$$

Now, we define a new coefficients:

$$\text{Let } \widehat{\phi}_{ij}(x) = \phi_{ij}(x) - \frac{\phi_{1i}(x)\phi_{1j}(x)}{\phi_{11}(x)} \quad \text{for } i,j > 2. \quad (3.113)$$

Then, we have:

$$\phi(x') = \phi_{11}(x)x_1'^2 + \sum_{i,j=2}^{n}\widehat{\phi}_{ij}(x)x_ix_j. \quad (3.114)$$

After the base case, $\phi(x')$ has no cross-terms involving x_1' and is of the form 3.114. This matches the inductive hypothesis for $r = 2$.

ii. **Inductive Step ($r \geq 2$)**:

3.3.4 **General Expression:** Assume we already diagonalize the first $r-1$ variables, so:

$$\phi(y) = \sum_{j=1}^{r-1}\pm y_j^2 + \sum_{i,j\geq r}^{n} y_iy_j\widehat{\phi}_j(y). \quad (3.115)$$

Just like how we isolated x_1^2 by eliminating x_1x_j terms, we now isolate y_r^2 by eliminating y_ry_j terms. This reinforces the inductive strategy.

$$\sum_{i,j\geq r} y_i y_j \widehat{\phi}_{ij}(y) = \underbrace{\widehat{\phi}_{rr}(y) y_r^2}_{\text{① pure } y_r^2} + \underbrace{2\sum_{j>r} \widehat{\phi}_{rj}(y) y_r y_j}_{\text{② cross-term}} + \underbrace{\sum_{i,j>r} \widehat{\phi}_{ij}(y) y_i y_j}_{\text{③ remaining terms}} . \tag{3.116}$$

Now, we define new coordinate y'_r:

$$y'_r = \sqrt{|\widehat{\phi}_{rr}(y)|}\left(y_r + \sum_{j>r} \frac{\widehat{\phi}_{rj}}{\widehat{\phi}_{rr}(y)} y_j\right) \implies y_r = \frac{y'_r}{\sqrt{|\widehat{\phi}_{rr}(y)|}} - \sum_{j>r} \frac{\widehat{\phi}_{rj}(y)}{\widehat{\phi}_{rr}(y)} y_j. \tag{3.117}$$

The scaling by $\sqrt{|\widehat{\phi}_{rr}(y)|}$ ensures the quadratic term $y_r'^2$ has coefficient ± 1, while the linear shift eliminates cross-terms.

3.3.5 **Analysis of the Terms:**

① Pure y_r^2 :

$$\begin{aligned}
\widehat{\phi}_{rr}(y) y_r^2 &= \widehat{\phi}_{rr}(y)\left(\frac{y'_r}{\sqrt{|\widehat{\phi}_{rr}(y)|}} - \sum_{j>r} \frac{\widehat{\phi}_{rj}(y)}{\widehat{\phi}_{rr}(y)} y_j\right)^2 \\
&= \widehat{\phi}_{rr}(y)\left(\frac{y'_r}{|\widehat{\phi}_{rr}(y)|^2} - 2\frac{y'_r}{\sqrt{\widehat{\phi}_{rr}(y)}} \sum_{j>r} \frac{\widehat{\phi}_{rj}(y)}{\widehat{\phi}_{rr}(y)} y_j + \left(\sum_{j>r} \frac{\widehat{\phi}_{rj}(y)}{\widehat{\phi}_{rr}(y)} y_j\right)^2\right) \\
&= \operatorname{sgn}(\widehat{\phi}_{rr}(y)) y_r'^2 - 2y_r^2 \sum_{j>r} \frac{\widehat{\phi}_{rj}(y)}{\sqrt{|\widehat{\phi}_{rr}(y)|}} y_j + \sum_{j,k>r} \frac{\widehat{\phi}_{rj}(y)\widehat{\phi}_{rk}(y)}{\widehat{\phi}_{rr}(y)} y_j y_k.
\end{aligned} \tag{3.118}$$

② Cross-Terms:

$$\begin{aligned}
2\sum_{j>r} \widehat{\phi}_{rj}(y) y_r y_j &= 2\sum_{j>r} \widehat{\phi}_{rj}(y)\left(\frac{y'_r}{\sqrt{|\widehat{\phi}_{rr}(y)|}} - \sum_{k>r} \frac{\widehat{\phi}_{rk}(y)}{\widehat{\phi}_{rr}(y)} y_k\right) y_j. \\
&= 2\sum_{j>r} \frac{\widehat{\phi}_{rj}(y)}{\sqrt{|\widehat{\phi}_{rr}(y)|}} y'_r y_j - 2\sum_{j,k>r} \frac{\widehat{\phi}_{rj}(y)\widehat{\phi}_{rk}(y)}{\widehat{\phi}_{rr}(y)} y_j y_k.
\end{aligned} \tag{3.119}$$

Now, we combine all the terms:

$$\begin{aligned}
\widehat{\phi}_{rr} y_r^2 + 2\sum_{j>r} \widehat{\phi}_{rj}(y) y_r y_j &= \operatorname{sgn}(\widehat{\phi}_{rr}(y)) y_r'^2 + \left(\sum_{j,k>r} -2\sum_{j,k>r}\right) \frac{\widehat{\phi}_{rj}(y)\widehat{\phi}_{rk}(y)}{\widehat{\phi}_{rr}(y)} y_j y_k \\
&= \operatorname{sgn}(\widehat{\phi}_{rr}(y)) y_r'^2 - \sum_{j,k>r} \frac{\widehat{\phi}_{rj}(y)\widehat{\phi}_{rk}(y)}{\widehat{\phi}_{rr}(y)} y_j y_k.
\end{aligned} \tag{3.120}$$

Overall, we have:

$$\phi'(y) = \sum_{j=1}^{r-1} \pm y_j^2 + \operatorname{sgn}(\widehat{\phi}_{rr}(y)) y_r'^2 + \sum_{i,j>r} \left(\widehat{\phi}_{ij}(y) - \frac{\widehat{\phi}_{ri}(y)\widehat{\phi}_{rj}(y)}{\widehat{\phi}_{rr}(y)} \right) y_i y_j. \tag{3.121}$$

After substitution, the function becomes:

$$\phi(y') = \sum_{j=1}^{r} \pm y_j'^2 + \sum_{i,j>r} \widehat{\phi}'_{ij}(y) y_i y_j. \tag{3.122}$$

where $\pm y_r^2$ has the sign of $\widehat{\phi}_{rr}(0)$, and the near coordinates are:

$$\widehat{\phi}'_{ij}(y) = \widehat{\phi}_{ij}(y) - \frac{\widehat{\phi}_{ri}(y)\widehat{\phi}_{rj}(y)}{\widehat{\phi}_{rr}(y)} \quad \text{for } i, j > r. \tag{3.123}$$

Now, comibining the base case and the inductive case, we now finished the proof of the **Morse's Lemma, Ref: 3.3**.

□

Summary: Combining Steps 1–3, the oscillatory integral $I(\lambda) = \int_{\mathbb{R}^n} e^{i\lambda\phi(x)}\psi(x)\,dx$ transforms via Morse's lemma (Step 3) to $\int e^{i\lambda Q(y)}\tilde{\psi}(y)\,dy$, where $Q(y)$ is a quadratic form and $\tilde{\psi} \in C_c^\infty$. Step 1 establishes the $\lambda^{-n/2}$ scaling for the pure quadratic phase, while Step 2 extends this to general test functions through decay estimates. The asymptotic coefficients a_j are determined by $\psi(x_0)$ and the Jacobian of the coordinate transformation that reduces ϕ to $Q(y)$.